

R&D, Spillovers, and Intellectual Property Rights

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Abstract

This paper examines the relationship between private R&D and intellectual property rights (IPRs). The relationship between these two variables is central to the debate on whether to strengthen or reform intellectual property laws and regimes. Do IPRs enhance the returns to R&D (and to what extent) or do they restrict knowledge diffusion (and to what extent)? Empirical evidence on these questions is currently lacking. This paper develops an optimizing, dynamic equilibrium model of R&D behavior, and estimates the model using OECD data on 21 countries and 18 manufacturing sectors over the period 1987-95. The model is estimated for the whole sample as well as by individual sectors. Furthermore, different kinds of IPRs are examined: patent rights, copyrights, trademark rights, software rights, parallel importation rights, and enforcement effectiveness. The results indicate that IPRs are a diverse concept and should not all be treated the same. Overall, the evidence favours the view that patent protection and enforcement stimulate private R&D -- largely indirectly, however, by stimulating the stock of private R&D spillovers. Copyrights, trademark rights, and parallel importation rights are generally found to have negative effects on private R&D. Software rights have positive, but secondary, influences on private R&D.

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1. Introduction

The focus of this paper is the relationship between private R&D and intellectual property rights. There is a common belief that market failures exist in R&D, due to the public good characteristics of knowledge and externalities associated with R&D activities. One solution is to have public sector funding and/or performance of R&D. Another is to foster private markets for R&D by creating private ownership or property rights over the products of R&D. As of yet, there are few, if any, studies that evaluate the success of intellectual property rights in stimulating R&D activities. The focus in previous work has been on the relationship between public R&D and private R&D.¹

The purpose of this paper is to help fill this gap in the literature, and examine the extent to which intellectual property rights (IPRs) explain private R&D capital. Moreover, the paper will not treat IPRs as some homogenous measure (which they are not), but will examine the role of different kinds of IPRs; for example, patent protection, copyrights, trademark rights, parallel import protection, software protection, and enforcement effectiveness. There are different views as to the ultimate effect of IPRs, which this paper will try to address; namely, whether they enhance R&D capital accumulation or restrict R&D knowledge diffusion. These different viewpoints are reflected in recent debates and will help shape the direction of reform in intellectual property regimes across countries. International IP reform involves not only the laws of developing nations but also those of industrialized countries that are launching new initiatives in new technological spheres (e.g. biotechnology, software, internet commerce, database protection, finance, and so forth).

This paper develops quantitative measures of different kinds of IPRs and empirically analyzes the extent to which they determine R&D capital stocks and R&D spillovers (or knowledge diffusion). Despite the lack of empirical studies (on the relationship between R&D and IPRs), there is no shortage of theoretical and policy debate. The proponents of stronger IPRs generally emphasize the potential effect on stimulating R&D as one of the benefits. The opponents emphasize the costs of stronger IPRs, which include limiting knowledge diffusion and frustrating the R&D of those who do not own the property right (for example, the R&D of rivals and follow-on inventors). Thus the impact of IPRs on R&D is to date an unresolved issue.

The outline of this paper is as follows: the next section provides a brief literature review of the few studies that investigate the impact of IPRs on R&D. Section 3 develops a model of IPRs and research and development. It derives the R&D investment equation that will be estimated. The fourth section discusses indexes of different kinds of IPRs. The fifth section discusses the data set of 21 countries and 18 manufacturing industries. The sixth section presents the empirical results. The last section (section 7) summarizes the main results and suggests extensions for future work.

2. Brief Literature Review

A number of empirical studies exist on the determinants of R&D behavior, but none focuses on the role of IPRs, at least not directly. The emphasis thus far has been on the role of public R&D,

¹ See, for example, David and Hall (2000) for a survey and analysis.

subsidies, tax credits, market size, firm size, corporate profits, concentration ratios, among others.²

As to the theoretical literature, conclusions about the welfare and efficiency effects of stronger intellectual property rights are quite mixed. For example, Takalo and Kanninen (2000) find that a strengthening of patent rights can delay the introduction of a new technology to the market (i.e. raises the value of the innovator to wait); Bessen and Maskin (2000) develop a model of sequential and complementary innovation in which patent protection reduces innovation and social welfare. Helpman (1993) in an international (North-South) setting argues that weak IPRs in the South may actually be welfare-enhancing for the South and that stronger IPRs in the South may not necessarily benefit the North (consumers, for example, would forgo the benefits of cheaper imitated imports). On the other hand, theoretical studies by Diwan and Rodrik (1991) and Taylor (1994) find that stronger IPRs may enhance global welfare, innovation, and productivity.

Though not always explicit about it, a large number of theoretical studies on IPRs are actually about patent rights and inventive activity. Landes and Posner (1987 and 1989) explicitly provide a theoretical analysis of non-patent IPRs, such as trademarks and copyrights. Essentially trademark protection encourages economic efficiency by reducing the search costs of consumers (by allowing them to recognize quality products by symbols or name). Furthermore, firms or intellectual property owners in turn invest in promotional expenditures to attract consumers and expenditures to maintain the quality of their products or services. Without being able to link their investments and products to their trademark, they would have less incentive to invest in those quality-promoting investments. There are occasions, however, where trademark protection can be too broad (e.g. a name or symbol becomes generic) and would increase the cost of business for rival firms such that economic efficiency is harmed in the aggregate. Copyrights over original and derivative works also stimulate creativity by increasing the odds of appropriating the benefits of the creations. Copyrights can also complement other rights, such as patent rights, where the ideas are not protectable but the expression is; for example, pure computer and mathematical algorithms. There may also be situations where stronger copyrights may be adverse to economic efficiency - namely where the rights reduce the incentive of rivals to create, or the owner's incentive to produce future new creations. Each creator is part of an intertemporal chain of creators. Thus stronger protection on expressions affect subsequent generations of creators (who themselves would like to build on previous works).

In terms of empirical work, a survey by Levin et. al. (1987) of U.S. firms finds that patent protection is not the most important means firms have to appropriate the returns to their R&D (as compared to lead time and reputation). Moreover, firms patent for reasons other than to protect their innovations (for example, to acquire strategic bargaining chips for cross-licensing negotiations). These findings suggest that patent rights are not very important to stimulating innovation. On the other hand, case studies conducted in developing countries indicate that IPRs are considered very important to innovation (see Sherwood (1990)). This suggests that the marginal value of patent rights (or IPRs) is higher in developing markets (where legal and other institutions are not as well developed and where, as a result, firms have few other alternative means of appropriation, if any). Another

² See David, Hall, and Toole (2000) for a survey.

interesting case study is by Korenko (1999) which finds that, for the Italian pharmaceutical industry, a strengthening of local intellectual property rights helped expand domestic R&D and market share (rather than result in a situation where foreign firms crowded out domestic).

3. Conceptual Framework

The following is an optimizing model of the demand for and supply of R&D capital.³ On the buyer's side, there exists a single aggregate representative firm that behaves competitively which, in order to produce output, purchases R&D inputs from private producers to be used along with other factors of production: labor and capital. Its output also depends upon the stock of public (government-financed) R&D capital. On the supply side, there are many firms producing R&D output for the demand side of the market. A single representative firm could have been posited, with no qualitative change in results (and indeed symmetric identical suppliers will be assumed shortly), but this modelling approach allows us to see how interfirm R&D spillovers can occur. The producers of R&D output solve an (almost) standard profit maximization problem, choosing their level of output to maximize profits; however, the cost of producing R&D output depends not only on the level (or quantity) of R&D output, but also on the stock of spillover R&D capital available to the firm. This feature captures the idea that there exist intertemporal knowledge spillovers. The marginal cost of producing R&D is lowered by a larger stock of past R&D available in the economy. The more knowledge there is, the less costly it is (holding other factors constant) to produce a given increment of knowledge. This helps to capture the "standing on the shoulders of giants" effect. Indeed the firm benefits not only from having done more R&D in the past but also from other firm's having done more R&D in the past. Thus intertemporal spillovers derive from both own R&D and spillover R&D. How easily available or accessible other firm's R&D knowledge is for the use of a firm depends, among other things, on the degree of substitutability between R&D performed by different firms and on whether or not there are technical and legal barriers affecting the transmission of knowledge from one firm to another (or from one sector to another, or one country to another).

This brings us to the question of how intellectual property rights play a role in the generation of R&D output. On the one hand, private producers can better appropriate the returns to their R&D efforts. The R&D is sold to the prospective users of R&D capital. With imperfect appropriability, the producers only capture a fraction of the potential sales to the users (or buyers). Imitators capture the remaining fraction of sales. Either way, imperfect IPR is modelled as reducing the revenues to the supplier. On the other hand, IPRs may be one type of legal barrier affecting interfirm spillovers. To the extent that intellectual property protections reduce knowledge diffusion, the degree of spillovers will be smaller, making it more difficult for firms to acquire spillover knowledge or enjoy the advantages of standing on the shoulders of previous researchers.

³ For non-optimizing equilibrium models of R&D expenditure behavior, see Lichtenberg (1987) and Howe and McFetridge (1976). The underlying basis is similar: investment in R&D proceeds to the point at which the marginal rate of return to R&D equals the marginal cost of funds.

Of course, there is one difficulty with this argument about IPRs inhibiting knowledge diffusion. What IPRs actually are intended to do is to exclude non-rights owners from practising or exploiting the new ideas (inventions, creations, and so forth). But IPRs do not preclude others from benefitting from the knowledge per se, as long as they apply the knowledge in some alternative way or produce a *differentiated* idea, concept, or expression. Indeed, an objective of having IPRs (particularly in the case of patent rights) is to encourage the rights-owner to disclose or reveal their underlying technology (sufficiently to enable persons reasonably skilled in the practice to replicate or duplicate the new idea). Thus, often the (temporary) intellectual property protection is given explicitly in exchange for the disclosure of new knowledge. In the case of patents, the disclosure is provided in the patent application itself and is available to the general public in print form, online, or other media. Thus, far from concealing knowledge, the objective of IPR laws is actually to enhance it. One therefore has to distinguish between technology diffusion (of intellectual *products* or *services* in the marketplace) from knowledge diffusion.⁴

Still, a case could be made that while knowledge diffusion itself is not affected, the manner in which knowledge can be used in research activity is affected by the laws and protections granted to others. The essence of spillovers is that research done by others has some *beneficial* effect (i.e. some positive externalities) on an agent. But the benefits may largely be derived only if the agent is able to exploit it in some fashion or other, and that certain laws may prohibit that exploitation. In this conceptualization, the enjoyment of spillovers requires some *learning by doing* or learning by producing. Tighter intellectual property protection may restrict an agent's ability to derive spillover benefits precisely because IPRs limit the manner in which non-rights holders may use or apply any proprietary knowledge. Thus, as long as learning by doing is an important means by which research spillovers can take place, then IPRs may weaken the beneficial (spillover) effects of knowledge diffusion. Under this view, knowledge diffusion or disclosure occurs but the economic impacts of that diffusion can be limited.⁵

⁴ Another false argument is that IPRs (particularly patent rights) impede knowledge diffusion because private agents will acquire proprietary rights over basic knowledge, which is vital to furthering knowledge. The argument is incorrect because technically patent rights are not granted to basic scientific and mathematical knowledge. Proprietary rights can be obtained for applied work, not theoretical work. This is not to say that in practice proprietary rights have not been given to what many would consider “theoretical” discoveries (for example, “expressed sequence tags” in genome research), but those patent grants would be considered granted in “error” and grounds for invalidation would exist.

⁵ This is not to say that the knowledge diffusion is limited to spillovers of technological knowledge between firms. There may, for instance, be useful spillovers for academic research. Moreover, even if no technological spillovers occur, there may be some valuable signals. Firms are provided with information on which technological areas have been covered (or solved) and which are open for discovery or exploitation (or further research). Intellectual property disclosure also has other functions. The patent is not only a scientific document but also a legal document. It shows what rights and intellectual claims the holder has. It is a document for

The following is a list of notation to be used:

IPR	Level of Intellectual Property Right
R_p	Stock of Private R&D Capital
R_g	Stock of Public R&D Capital
I_p	Investment (Flow) in Private R&D Capital (and Measure of Private R&D Output)
I_g	Investment (Flow) in Public R&D Capital
K	Stock of Physical Capital
L	Labor
Y	Output
V	Value of Firm
	Profits
i	Investment in Private R&D Capital by the i th-firm
r_i	Stock of Private R&D Capital of the i th-firm
R_i^*	Stock of Spillover Private R&D Capital of the i th-firm
N	Number of R&D-Producing Firms
c	Cost of Producing Private R&D Output
p_R	Price of Private R&D Output
	Geometric Depreciation Rate of R&D Capital
	Real Interest Rate
	Degree of Substitutability between Own R&D and Spillover R&D
	Degree of Appropriability or Market Share of Intellectual Property Rights Holder
	Degree of R&D Spillovers derived by firms

In what follows, the optimizing demand for R&D will first be characterized, followed by the optimizing supply of R&D. The two equations are then solved to yield the steady-state stock of R&D capital. On the demand side, the representative aggregate firm maximize the following functional:

$$(1) \quad \max_{\dot{R}_p, R_p} V = \int_t^{\infty} [Y(R_p, R_g, \dots) - p_R I_R] e^{-\int_t^s \rho_u du} ds$$

subject to:

$$(2) \quad \dot{R}_p = I_p - \delta R_p$$

where output is a function of the stocks of private and public R&D capital. (For the moment, we

purposes of legal enforcement. Hence, another kind of knowledge disclosure is knowledge about the market structure, its organization, niche, and positioning of competitors, if any.

suppress the other factors of production, such as physical capital K and labor L.) The following first and second partial derivatives of the production function are assumed (although the ‘signs’ of these derivatives are the subject of empirical debate): $Y_1 > 0$, $Y_2 > 0$, $Y_{11} < 0$, and $Y_{22} < 0$.

Public R&D investment, I_g , is treated as exogenous here, and public R&D capital evolves as follows:

$$(3) \quad \dot{R}_g = I_g - \delta R_g$$

The necessary condition for value maximization is:

$$(4) \quad Y_1 = (\rho + \delta) p_R - \dot{p}_R$$

This is the standard optimality condition where the LHS is the marginal product of R&D capital ($\partial Y / \partial R_p$) and the RHS the user cost of R&D capital.

On the supply side, assume N producers of R&D output. The i th firm (where $i = 1, \dots, N$) chooses its level of R&D output, v_i , to maximize the present discounted flow of profits:⁶

$$(5) \quad \max_{v_i} \Pi_i = \int_t^\infty [\theta p_R v_i - c(v_i, R^*)] e^{-\int_t^s \rho_u du} ds$$

subject to

$$(6) \quad \dot{r}_i = v_i - \delta r_i$$

where p_R represents firm revenues, of which a fraction θ is actually appropriated by the firm (due to imitation). It is assumed that $\theta = (1 - \beta)$ (IPR), where the partial derivative $\beta' > 0$ and $0 \leq \beta \leq 1$. In the absence of imitation, $\beta = 1$; under perfect imitation, $\beta = 0$.

The interpretation here is that θ is the market share of the intellectual property rights holder (and $(1 - \theta)$ the share of imitators). Both the rightful owner and imitators compete and charge the same price p_R to the user of R&D and produce identical outputs.⁷ (Of course, equilibrium p_R could be higher (or lower) the fewer (or more) imitators there are). At that price, if I_p denotes the total R&D output

⁶ In this specification, neither the variety nor the quality of products resulting from R&D (or inventive) activity is explicitly treated.

⁷ To focus on the main qualitative results of this paper, the behavior of imitators is not explicitly specified. One can assume some simple constant marginal cost structure and free entry/exit among them. We also do not model the behavior of inventors and innovators in the background, and omit the details of how R&D producers obtain the exclusive right (or exclusive license) to manufacture R&D output. This too would detract from the paper's main focus.

demand, the composition of output produced between the rights holders and imitators are as follows:

$$I_p = I_p + (1 - \alpha)I_p, \text{ where } I_p = I_i, \text{ for } I = 1, \dots, N.$$

In (5), $c(\cdot, \cdot)$ is the cost function for R&D output. The cost of producing R&D output depends positively on the quantity of output produced, Y , and negatively on the stock of spillover R&D knowledge, R^* , where

$$(7) \quad R^*_i = (r_i + \psi \sum_{j \neq i}^N \omega_j r_j)$$

and

$$(8) \quad R_p = \sum_{i=1}^N r_i$$

That is, the stock of R&D generates intertemporal (including inter-firm) externalities. Firm's investments (or experiences) in R&D generate some learning by doing externalities (which affect the future marginal cost of R&D production). It is assumed that $c_1 > 0$, $c_{11} > 0$, $c_2 < 0$, $c_{22} > 0$, and $c_{12} < 0$. Thus, an increase in the stock of spillovers reduces the marginal cost of R&D production but at a diminishing rate.

In (7), R^*_i is the spillover R&D available to the i th firm. It is composed of the firm's own past R&D stock (that is, the knowledge capital that it created) and of other firms' R&D stock. This may include the R&D of firms in other sectors (or other countries). ψ measures the degree of substitutability between one firm's R&D and that of another. It may also be referred to as the measure of technological similarity between sectors. It is likely that the R&D of other firms would be less appropriate to the research activity of a given firm, yet it is difficult to measure exactly how substitutable research activities are among different firms within or between different industries, particularly in situations where technological frontiers evolve in various directions. In some cases, no usefulness of research might be expected between firms in diverse sectors, yet some spillovers have been generated (e.g. between semiconductor research and quartz watch research, or heart surgery research and automotive research). For simplicity, since ω_j cannot be measured or estimated at the level of aggregation we have in this paper, we shall assume throughout that $\omega_j = 1$ for all j .

As discussed earlier, R&D spillovers from other firms to the i th firm may be "blocked" by the intellectual property rights of other firms which prohibit the i th firm from utilizing the knowledge of others, thereby limiting its ability to capture R&D spillovers. α measures the extent to which external R&D "spills" in, where it is assumed that $\alpha \in [0, 1]$ (IPR), and the partial derivative $\partial c / \partial \alpha < 0$ and $0 \leq \alpha \leq 1$. Where IPRs do not prohibit spillovers at all $\alpha = 1$, and where IPRs completely shut off spillovers, $\alpha = 0$. Note (in (7)) that the variable α does not affect the i th firm's ability to enjoy spillovers from its own past R&D, r_i .

Before characterizing the optimality conditions for the supplier (i.e. the solution to (5)), it is worth going over some special cases:

Case 1: No Intertemporal Spillovers

In this case, the cost function for producing R&D output is just $c = c(i)$, and the necessary condition for profit maximization is the standard condition:

$$p_R = c'(i)$$

whereby the level of R&D output, i , is chosen at which the price of R&D output (adjusted for the degree of appropriability of revenues) equals the marginal cost of production. Inverting the necessary condition gives $i = i(p_R)$. Since all firms face the same price and if they also have identical cost functions, $i = i$; that is, they will produce the same amount of R&D output. Hence aggregate R&D output (or aggregate investment from the point of view of the demand side) is:⁸

$$I_p = I_p(p_R) = N(i(p_R))$$

The supply of R&D is positively related to the price of R&D capital, which in steady-state is (using equation (4)) $p_R = Y_1 / (\delta + \rho)$. Thus in steady-state, $I_p = I_p(Y_1 / (\delta + \rho))$; that is, a function of the discounted marginal productivity of private R&D capital, adjusted for the degree of appropriability. In this case, an increase in the stock of private R&D capital would depress investment due to the diminishing marginal productivity of R&D capital. And tighter intellectual property protection, to the extent that it raises the degree of appropriability of R&D revenues, ρ , would stimulate R&D investment. In the next two cases, these two predictions are ambiguous.

Case 2: Small Firm Case -- Exogenous Spillovers

In this case, in definition (7), r_i is considered sufficiently small relative to the stocks of R&D of other firms that the i th firm treats the stock of (economy-wide) spillovers R^* as given. The firm would not take into account the contribution of its R&D output to future cost reductions. The cost of producing R&D output decreases with the stock of R&D spillovers (or with society's stock of R&D), but the firm's own output constitutes a negligible share. The effect of this (compared to the next case) is that the firm's problem remains "static" as in Case 1 above; that is, the firm chooses output to maximize instantaneous profits. The necessary condition for profit maximization is:

$$p_R = c'(i, R^*)$$

(In the special case where $\beta = 1$, R^* simply coincides with the aggregate stock of private R&D capital, R_p .) Combining the above condition with the optimizing condition from the demand side (4), aggregate R&D investment, I_p , will be found in steady-state equilibrium to have an ambiguous relationship with the aggregate stock of private R&D and with level of IPR. A higher stock of aggregate private R&D, on the one hand, reduces the cost of producing an additional unit of R&D

⁸ More precisely, $I_p = N \cdot i = N \cdot (i(\cdot) + (1 - \beta) \cdot i(\cdot))$, if we take into account the imitators.

output; on the other hand it reduces the marginal productivity of R&D capital (and reduces the market's demand for R&D output). A higher IPR level increases the appropriability of R&D investment on the one hand and may (on the other hand) reduce the transmission of spillovers among firms. Thus the overall impact on equilibrium R&D is ambiguous.

Case 3: Large Firm Case -- Endogenous Spillovers

In this case, we allow for the possibility that the R&D supplying firm is large enough to realize that its own production of R&D helps to reduce its future marginal cost of R&D production.⁹ The steady-state comparative statics effects of a change in the aggregate stock of private R&D and in the level of IPR are the same (as in Case 2). Indeed, one purpose of comparing cases 2 and 3 is to show that the ambiguous effects of IPR and the stock of private R&D capital on R&D investment do not depend on the assumption of small vs. large firm. However, both the dynamics and the steady-state equilibrium levels of R&D are affected by the nature of the firm (and how it perceives its stock of R&D vis-a-vis that of others).

For the i th firm, the solution to maximizing (5) s.t. (6) (via the choice of $r, \dot{r}$) is:

$$(9) \quad \dot{z}_i = (\rho + \delta)z_i + c_2$$

$$(10) \quad z_i = c_1 - p_R$$

where $c_2 = \partial c / \partial R_i^*$ is the marginal reduction in cost due to R&D spillovers, and $c_1 = \partial c / \partial \dot{r}_i$ is the marginal cost of producing R&D. z is the excess of the marginal cost of producing R&D over the marginal benefit of producing R&D. $z > 0$ as long as spillover research and development has a beneficial “negative” effect on the marginal cost of producing R&D. The firm is led to produce beyond the point where the marginal cost equals the marginal benefit of producing R&D. Figure 1 illustrates the optimal R&D output and the size of z . Figure 2 shows how the equilibrium R&D output increases with a larger stock of spillovers.

With symmetry, $\dot{r}_i = \dot{r}$, $r_i = r$, and thus $R^* = (1 + (N-1))r$. Now, in *steady-state*, using (2)-(4), (6), (8)-(10), we obtain:

$$(11) \quad c_1 \left(\frac{\delta R_p}{N}, (1 + \psi(N-1)) \frac{R_p}{N} \right) + \frac{c_2 \left(\frac{\delta R_p}{N}, (1 + \psi(N-1)) \frac{R_p}{N} \right)}{\rho + \delta} = \frac{\theta Y_1(R_p, R_g)}{\rho + \delta}$$

⁹ Its own R&D also contributes to the stock of R&D spillovers of other firms (as theirs contribute to this firm's spillover R&D stock). The firm, however, does not take into account the benefits other firms derive when choosing how much R&D to produce. Overall, the equilibrium amount of R&D should be lower than it would otherwise be. It would be a useful extension to develop a model and data set that would help determine the extent to which underinvestment in R&D occur (vis-a-vis a social planner outcome).

Totally differentiating (11) with respect to R_p , R_g , and IPR yields:

$$(12) \quad dR_p = \Omega_g dR_g + \Omega_I dIPR,$$

where

$$(12a) \quad \Omega_g = \frac{\frac{\theta Y_{12}}{\rho + \delta} + \delta(c_1 + \frac{c_2}{\rho + \delta})}{\Omega}$$

$$(12b) \quad \Omega_I = \frac{\frac{\theta' Y_1}{\rho + \delta} - (N - 1)\psi/r(c_{12} + \frac{c_{22}}{\rho + \delta})}{\Omega}$$

$$(12c) \quad \Omega = \frac{1}{N}[\delta(c_{11} + \frac{c_{21}}{\rho + \delta}) + (1 + (N - 1)\psi)(c_{12} + \frac{c_{22}}{\rho + \delta})] - \frac{\theta Y_{11}}{\rho + \delta} > 0$$

Thus $\Omega_I (=dR_p/dIPR)$ measures the comparative statics effect of a change in the IPR level on the steady-state stock of private R&D. The coefficient indicates what the effect depends on. The first term is positive and the second term negative. It is assumed that in absolute value $|c_{12}| < |c_{22}|$, implying that, across steady-states, the stock of spillover R&D reduces the marginal cost of producing R&D (even after allowing for the fact that the steady-state level of R&D produced will be higher). The first term measures the impact of IPR on the appropriability of R&D returns; the second measures the extent to which IPR “limits” (if at all) spillovers from occurring. The second term would vanish if $\psi = 0$ (meaning that external R&D is not appropriate to the firm). The effect of having more firms (i.e. having larger N across steady-states) is also ambiguous: more firms could mean more potential spillovers or more spillovers forgone due to any restrictions imposed by IPRs.¹⁰ Note that in (12c), Ω is positive as long as the overall (steady-state) cost of producing R&D rises with the stock of private R&D (even after netting out the effect of intertemporal and interfirm spillovers). Otherwise, the equilibrium stock of private R&D would approach infinity.

A few words on the dynamics. The dynamics of the system are governed largely by equations (4), (9), and (2). Combining (4) and (9) yields an equation of motion for the marginal cost of R&D output, c_1 . Along the equation where $\dot{c}_1 = 0$, the marginal cost and the stock of private R&D vary inversely. Holding the marginal cost constant, across steady-states, a higher stock of private R&D is associated with lower marginal returns to R&D capital (Y_1) and lower marginal returns to spillover R&D (since spillover R&D reduces the marginal cost of R&D production at a decreasing rate, since $c_{22} > 0$). This would put pressure on the marginal cost of R&D production to increase. Thus, in order to keep c_1 constant, a lower marginal cost must be associated with a higher stock of private R&D.

¹⁰ It would be useful to test the role of N using data on number of establishments. At present the data are incompletely available. See the OECD's *Structural Statistics for Industry and Services (SSIS)*.

Now, R&D investment is a function of the marginal benefit of R&D, namely p_R , but this in turn equals the marginal cost of R&D, c_1 , in equilibrium. Hence investment in R&D is a function of c_1 ; more specifically a positive function. As Figure 3 indicates, marginal cost c_1 and investment I_p covary positively. Thus the equation of motion for the stock of R&D (equation (2)) is indirectly a function of c_1 . As shown in Figure 4, the equation for $\dot{R}=0$ is positively sloped. A greater stock of private R&D involves greater depreciation, and therefore requires a greater steady-state flow of investment. A higher steady-state investment in R&D is associated with a higher steady-state marginal cost of producing R&D. Figure 4 indicates also the saddlepath trajectory towards steady-state. Along this path, as the marginal cost falls (rises), the stock of private R&D rises (falls).

For purposes of empirical implementation of equation (11), one can either linearize around steady-state and estimate accordingly or work with some convenient functional forms. For example, we can assume Cobb-Douglas functions for output and cost, and some simple exponential functions for the variables that are influenced by IPRs:

$$(13a) \quad Y = A R_p^{\alpha_1} R_g^{\alpha_2} K^{\alpha_3} L^{\alpha_4}$$

$$(13b) \quad c = B \psi^{\beta_1} R^{\beta_2}$$

$$(13c) \quad \theta = IPR^{\mu}$$

$$(13d) \quad \psi = IPR^{-\sigma}$$

where $A > 0$, $B > 0$ are constants. Note that we explicitly included other factors of production in the production function; namely, capital K and labor L .

Substitute (13a-d) into (11), and taking logs yields:¹¹

$$(14) \quad \ln \left(\frac{R_p}{L} \right) = \phi_0 + \phi_1 \ln \left(\frac{R_g}{L} \right) + \phi_2 \ln \left(\frac{K}{L} \right) + \phi_3 \ln (IPR) + \epsilon$$

where

$$\phi_0 = \text{constant} = \ln \left[A(1 - (1/N))^{\alpha_2} / (B(\alpha_1 + \alpha_2)(1/N)^{\alpha_1}) \right] / (\alpha_1 - \alpha_2 - \alpha_1)$$

$$\phi_1 = \alpha_2 / (\alpha_1 - \alpha_2 - \alpha_1)$$

$$\phi_2 = (\alpha_1 - \alpha_2 - \alpha_1)$$

¹¹ In the derivation, we incorporated the fact that $\log x = \log(1 + x)$, for some variable x that is large.

$$\beta_3 = (\beta_1 - \beta_2) (\beta_1 - \beta_2 - \beta_1)$$

To derive (14), constant returns to scale were imposed in both the production and cost functions (i.e. that $\beta_1 - \beta_2 = 1$ and $\beta_1 + \beta_2 + \beta_3 + \beta_4 = 1$). Otherwise, an additional term (namely $\ln L$) would appear on the RHS. In preliminary empirical tests, the constant returns specification could not be rejected. Equation (14) is expressed in terms of aggregate quantities, not firm level quantities. As the empirical section explains, equation (14) is estimated using manufacturing sector data. The error term e can be due to random deviations from the steady-state conditions which yield this equation, or from multiplicative random shocks to the production function and/or cost function (see the A and B terms).

4. IPR Indexes

Thus far, few econometric work exists because, until recently, there have not been many measures -- or indexes -- of intellectual property rights. Patent rights indexes have been developed, but non-patent IPRs have not. This section therefore describes indexes of these other kinds of IPR.

In total, six different kinds of IPR indexes will be used in the empirical investigation. Three of them cover standard statutory rights: patent rights, copyrights, and trademark rights. Two of them deal with aspects of IPRs that have been the subject of much recent policy debate: software protection and parallel import protection. The last is a qualitative measure on the effectiveness of enforcement in practice. The previous five measures are considered largely to capture statutory strength, which is not to say that statutes do not play a role (as in providing signals about the intent of policy authorities). More on this issue later.

A. Preliminary Remarks

Multiple indexes of IPRs are examined in order to get a broad perspective on the state of IP rights in a country. Individual intellectual property owners (or potential owners) may be heterogeneous as to what kinds of IPRs they value. As an analogy, consider the surveys that rank cities according to quality of residential life (e.g. surveys on “best places to live”). What do people look for in a city: school quality, low crime, low taxes, scenic views, quality of air? How do people rank these different characteristics? Obviously residents would want all of these good characteristics, but how would they prioritize them? What weights would they attach to the different characteristics? Likewise, what do inventors, artists, writers, producers, and so forth, look for? Ease of application, no compulsory licensing, no working requirements, strong penalties for infringement, expansion of rights to cover new areas (e.g. software, biotechnology, internet commerce, folk dances, etc.)? Again, how would they prioritize and weight different IP law features? Of course, a major difference between rating IP systems and rating cities is that in the latter the surveys are attempting to measure something about “quality”. The rating of IP systems, in contrast, is not about measuring the quality of IP regimes, but rather the “strength” of IP regimes. It is not for instance attempting to determine the “optimal” level of protection. The optimal level need not be the one associated with maximal strength. Quality and strength may go together, but they are distinct. Issues of quality would deal

with the equity of rights (between different intellectual property owners, and between them and non-owners), the effect on welfare and economic efficiency. The indexes here measure how protective systems are of the rights of intellectual property owners.¹² The empirical section then determines whether certain efficiency factors (like productivity and innovation) are influenced by the “strength” of those IP rights.

Another important remark is that the indexes largely measure statutory levels of protection (i.e. of laws on the books) rather than actual practice, although this study does incorporate a few variables which help to assess the actual enforcement of laws. Nonetheless a common concern is that IP indexes only measure perceived protection - not actual. However, as will be discussed later, the correlation between statutory protection and actual enforcement, while not perfect, tends to be high. Countries that have strong laws on the books tend to be the ones that also actually carry out the laws. Moreover, independent of enforcement, statutes can play a role. For instance, the empirical results will show that even perceived (statutory) protection has real effects. This might be due to, among other things, a ‘signaling’ effect. The laws on the books may affect agents’ expectations or confidence levels, and thus affect their investment and other decisions.

Related to the issue about perceived vs. actual protection is the practice of judging the accuracy of index values according to certain a priori views. Of course, it is useful to incorporate information based on experiences and expert opinion. The indexes and ‘expert opinion’ should be viewed not as substitutes but as complements. However, a common pitfall is to judge whether a country’s IP index value is too low or too high according to the country’s level of economic development, the a priori assumption being that richer countries should have stronger levels of IP protection. In general this is the case, but there are instances where it is not (some rich countries have IP weak systems, and some poor economies have strong systems). In such cases, other factors are not held constant (for example, richer economies with weak IP systems may have good educational systems to compensate, or poorer economies with strong IP systems may follow poor fiscal and monetary policies which offset the effects of IPRs). In all cases, it should be understood that the IPR indexes are not measures of economic development. They may be important determinants of development, but are not themselves indicators of it. The approach in constructing these IP indexes should be to let the chips fall where they may, with minimal (if any) reliance upon a priori views about the economic consequences of IPRs.

Legal features are chosen which measure the “strength” of intellectual property regimes. If there is ambiguity or uncertainty as to whether a feature contributes to the strength of rights, it is not

¹² A further distinction might be made between incumbent vs. entrant. The indexes tend to measure the strength of existing rights holders. For example, if existing IP owners exercised a very broad scope over their rights, the laws may make it difficult for new inventors to obtain intellectual property protection. This might be interpreted as a failure on the part of the system to provide intellectual property rights for the entrant. This distinction, however, is not pursued here. Perhaps the solution is to develop different indexes for different classes of inventors or creators.

incorporated (for example, priority rules: first-to-file vs. first-to-invent). Another guiding principle in choosing legal features is not to be exhaustive but selective: that is, to choose those legal features that yield maximum variability across countries.¹³ Furthermore, the information has to be widely available across countries.¹⁴

B. Overview

The eight measures of IPR considered in this study cover the gamut of statutory and enforcement provisions, piracy and enforcement experiences.

For each of the three basic IP instruments (patents, copyrights, and trademarks), the index consists of four sub-categories: Coverage, Duration, Restrictions, and Membership in International Treaties. Enforcement can also be included as a sub-category (as it is in Ginarte-Park (1997)). However, since the enforcement provisions are available for the enforcement of all three types of rights (patents, copyright, and trademark), it can serve as a distinct index or be used in all three of the basic types of IP rights as a fifth category. Or it can be combined with the “qualitative” enforcement index.

Coverage refers to the subject material (type of invention, expression, or symbol) that can be protected; duration refers to the length of protection; restrictions refer to the less than exclusive use of those rights; membership in international treaties indicates the adoption into national law of certain substantive and procedural laws of those international agreements. Note that, for signatory nations, there may be some ‘double-counting’ in that a nation gets credit for having certain legal features in national law, but those features may be part of an international law treaty to which the nation is a signatory and for which the nation already gets credit for being a member of that particular treaty. However, membership in an international treaty in and of itself provides some value added information, particularly about the willingness of particular nations to adhere to shared international principles, such as non-discrimination.

The following sub-section provides further details about each of the measures of IP protection. The acronym to be used in the empirical section is given in parentheses. Appendix 1 provides a quick summary of the legal features in each type of IPR index and of how the indexes are scored.

C. Description of Indexes:

i) Patent Rights (PAT): The measure of patent rights is taken from Ginarte and Park (1997) and

¹³ For example, incorporating ‘derivative works’ adds no variability because all countries in the sample provide protection for them in their copyright laws.

¹⁴ For instance, only few countries specify the levels of punishment or penalties for IP violations (e.g. length of sentences, amount of fines, etc.). Most countries indicate that infringement can be punishable as a civil or criminal offense, but are not explicit enough to allow for comparisons of punishment levels across countries.

Park, Vijaya, and Wagh (1999).¹⁵ The index of patent rights ranges from zero (weakest) to four (strongest). The value of the index is obtained by aggregating the four sub-indexes: extent of coverage, membership in international treaties, duration of protection, and absence of restrictions on rights (such as compulsory licensing).

The numerical value of each sub-index ranges from zero to one and indicates the fraction of legal features in that sub-index available in the particular country. For example, a value of $\frac{1}{3}$ for membership in international treaties indicates that a country is a signatory to one-third of the international treaties listed under that sub-index. A value of $\frac{1}{2}$ for duration implies that a country grants protection for half the international standard time (of 20 years from the date of application or 17 years from the date of grant). The value for coverage indicates the fraction of invention classes the country allows as patentable subject matter. Finally, several conditions exist under which authorities can revoke or reduce patent rights. The value for the 'restrictions' category indicates the fraction of those restrictions which are not exercised in the country.

(ii) Copyrights (COPY): This index varies also from 0 to 4. Each of its four categories is scored out of one. The score is again the fraction of features that are available. The coverage category includes those works that are among the primary victims of piracy, such as literary, dramatic, artistic, musical, cinematographic works, and so forth. The duration of protection is based on an international standard of 50 years. Note that countries may provide different lengths of protection for different types of copyrightable works. The duration score for each of these types of works is the ratio of its statutory duration to 50 years. If more than 50 years of protection is provided, the maximum score of 1 is given. The country's overall duration score is the average of the duration scores of the different types of copyrightable works.

The category on 'restrictions' includes rights to resale (i.e. *droit-de-suite*) which permit the copyright owner to share in a percentage of all subsequent sales of her work, thus enabling her to benefit from any appreciation in the value of her works. It also covers extended collective licensing schemes. Collective Licensing Societies are organizations of authors and performers. These societies are somewhat common in Europe. Their extended licensing schemes are deemed to weaken copyrights since they can at times interfere with the freedom of contract of an individual rights holder (see Campbell and Cotter (1997)); moreover, the licensing schemes may typically permit more liberal reproduction of works by photocopy or by broadcasting. For instance, organizations that obtain authorization from a collecting society to photocopy some author's work may in some cases be entitled to photocopy that author's published works in the same field not represented by the collecting society. Also, authorization to record works in a broadcast may also include authorization to record works of non-represented rights holders that happen to be in the same broadcast.

The copyright restrictions category also incorporates compulsory licensing. One kind of compulsory licensing is for private use and another for government use. Private individuals may apply for a

¹⁵ The authors obtained information on national patent laws from Baxter (2000) and WIPO (2000).

compulsory licence in some jurisdictions if (typically) a foreign work is not available locally after it is published elsewhere for some specified period of time (e.g. six months or a year, etc.). Some countries provide explicitly for compulsory licensing (e.g. in the U.S. for satellite broadcasting) and mechanical licensing (e.g. for musical works, etc.). These are treated as for private use. Government use (e.g. by a Ministry of Culture) is typically for educational purposes, for local technological development, or for judicial and administrative uses (such as proof in legal proceedings).

Major international copyright treaties include the Berne Convention, Rome Treaty, Universal Copyright Convention (UCC), and Phonogram Convention. The Berne convention is the oldest international copyright treaty, providing for uniform global protection. The basic underlying principles are national treatment, automatic protection, and independence of protection (independent of whether protection exists in the country of origin of the work). The Rome Convention protects “neighboring rights” (rights of performers). The UCC provides minimum legal obligations for each contracting state, emphasizing rights and protections that ensure an author’s economic interest. The Phonogram Convention focuses on strengthening rights of phonogram producers (given the increased piracy of records and tapes, at the time of its signing). This Convention, unlike the Rome Convention, does not provide substantive rights; as long as phonograms are protected, the mode of protection is left to domestic law (see Leaffler (1997), p. 451).

(iii) Trademark (TMARK): The trademark index also varies from 0 to 4. It is the sum of scores from four categories (again coverage, duration, restrictions, and membership in international treaties). Each category is scored out of one (indicating the fraction of available provisions). The coverage category lists three types of marks: service marks, certification marks, and collective marks.¹⁶ Countries vary over whether these types of marks can be granted trademark protection. *Service marks* are words, names, symbols, or devices that identify services. *Certification marks* are words, names, symbols, or devices that certify the origin (e.g. region) of particular types of goods, such as Champagne. These marks help identify the type of product. *Collective marks* identify trade associations or membership in some cooperative or other organization. The association (or its independent members) may be responsible for some product(s). The collective mark should tie the product(s) to the reputation of the collective.

For the duration of trademark protection, the international norm is 10 years. Again, the duration score is ratio of the statutory length of protection to 10 years; if the statutory length exceeds 10 years, the maximum score of 1 was given. The restrictions category examines whether countries require proof of use at the time of trademark rights renewal (e.g. demonstrate commercial use); whether there are linking requirements (e.g. linking foreign trademarks to a locally-owned firm); whether there are licensing restrictions (on royalties, technology transfer agreements); and whether there are conditions for the protection of well-known marks (e.g. that they be used in the local economy).

¹⁶ Since all jurisdictions with trademark laws allow words, names, symbols, devices, or any combination, to be trademarked, it was not necessary to list these under the coverage category. In some cases, color, sounds, fragrances, or 3-dimensional objects, can be registered, but the cross-country variations in the protection of these are small.

The international treaties category includes three major treaties: the *Madrid Agreement* which governs the international registration of marks. It does not protect any trademark rights but facilitates the acquisition of trademark rights in member states. The *Nice Agreement* governs the international classification of goods and services for the purposes of registering trademarks. Official documents and publications refer to these classes. Countries however can use this international classification as their principal system of classification or alongside their own national classification system. The *Paris Convention* also has provisions on Trademark rights. The Convention deals with the use of registered marks and of well-known marks.

(iv) Parallel Import Protection (PARA): Parallel imports refer to the imports of legally manufactured products by agents other than those with exclusive distribution rights. The right to prevent parallel imports is essentially the international equivalent, or extension, of domestic vertical restrictions. Domestically, wholesalers may grant retailers exclusive dealerships to help solve certain free rider problems (whereby other distributors, such as parallel traders, free ride on the promotion and other marketing activities of authorized dealers).¹⁷

Countries also vary in how they treat parallel importation. Under a national exhaustion system, parallel imports are not permitted; under an international exhaustion system, they are; under a regional system, parallel importing is permitted within the region, but are not permitted from outside the region. Based on these different policy regimes, the *International Intellectual Property Alliance* has undertaken a survey of whether IPR owners can be protected against parallel imports. This index has three values: 1 if yes, 0 if no; ½ if probably yes.

(v) Software Rights (SOFT): In light of the prominent developments in the computer industry (particularly software) and the impact it has had on economies, it would be useful to incorporate the effects of software protection. The above measures do not explicitly, if at all, incorporate computer software in their coverage categories. A separate index would be required. Software protection is available in several forms, depending on whether it is the idea (technical effect) or expression that is sought for protection. Thus, agents can use either patents or copyrights, or a combination of them. Software can also be protected through existing trade secret laws. Hence this index is the sum of three components: trade secrecy, copyrights, and patents. Each of these components gets a score: 1 if such protection is available, zero otherwise. In the case of software patentability and software copyrightability, a ½ score is given if the protection is possibly available but is not clear from existing statutes. In total, this SOFTWARE index varies from zero to three.

(vi) Enforcement Provisions (ENF): This index is the fifth component in the Ginarte and Park (1997) index of patent rights, separated from the rest of the components. In this category, the selected conditions are the availability of: *preliminary injunctions*, *contributory infringement pleadings*, and *burden-of-proof reversals*. A country that provides all three receives a value of 1 for this category. While litigation, arbitration, and settlement comprise different enforcement ‘routes’ should infringement occur, patent holders may have recourse to a number of statutory provisions

¹⁷ For further discussions of the effects of parallel import protection, see Maskus (2000).

which can aid in enforcement. Preliminary injunctions, for example, are pre-trial actions that require the accused infringer to cease the production or use of the patented product or process during the course of the trial. Preliminary injunctions are a means of protecting the patentee from infringement until a final decision is made in a trial. Contributory infringement refers to actions that do not in themselves infringe a patent right but cause or otherwise result in infringement by others. Thus, contributory infringement permits third-parties also to be liable if they contribute negligently to the infringement. Burden of proof reversals put the onus on the accused to prove innocence. Given the difficulty IP owners may have of proving that others are infringing on their ideas, expressions, or symbols, the shift in burden can be a powerful enforcement mechanism.

(vii) Enforcement in Practice (USTR): At present no scientifically conducted studies have been done on how laws are actually enforced in practice. The closest available are reports filed with the *U.S. Trade Representative (USTR)* concerning intellectual property enforcement in various countries. A major limitation is that the reports are biased towards the views of U.S. firms (of what constitutes effective and adequate enforcement). Another limitation is that some complainants may have ulterior motives for filing complaints; for example, to seek assistance in penetrating foreign markets because they are not able to compete against local firms on price, product quality, or other factor alone. A third limitation is that because the reports are descriptive and qualitative in nature, any attempt to construct quantitative indexes based on them is likely to depend subjectively on the author's interpretation of the nature of complaints.

On the other hand, having no measure at all of enforcement in practice would be a serious omission. Thus, notwithstanding these limitations, an index is developed to reflect the experience of IP enforcement as documented in these reports (see *USTR National Trade Estimates*). The index can then be compared to, and used in conjunction, with the other, largely statutory, IP indexes.

The index focuses on the execution of laws. Laws may be ineffectively implemented: i) because of a *lack of willingness* on the part of policy authorities to provide or enforce them (because the authorities, for whatever reason, do not agree with a strong intellectual property policy), or ii) because of a *lack of capacity* to enforce laws effectively. This may arise because of a lack of resources, training, and experience.

IP violations occur not only because of weak laws and enforcement, but also because imitators or infringers are very capable of copying. Therefore it is important to control for the capacity of a nation's "imitative" sector to make copies. In nations where the capacity for imitation is low, weak enforcement may not be an important factor to innovators. The weak capacity for imitation itself acts as a protection against imitation. On the other hand, even if strong laws exist (on the books) and enforcement is strong (that is, the authorities are both willing and able to protect rights), there will always be some infringement (even in regions where IP laws and rights are strong, such as in the U.S.). Thus the level of infringement activity is not in and of itself a good indicator of whether the laws are lax or ineffective, particularly if the laws are there and the court system enforces them (in which case the system is working well). While lax laws and poor enforcement do contribute to IP infringement, there are other factors that drive IP infringement activity (including capacity for

imitation, such as the level of technology (for copying), and quantity of innovations and creations).

Thus, for purposes of this index (which is to try to measure the actual enforcement of IP laws), the focus will be on how the authorities enforce or carry out the laws in practice -- not on the actual extent of infringement activity. This particular index looks first for whether enforcement mechanisms are available or adequate; secondly, whether laws are enforced; and thirdly, how effectively. For instance, if enforcement measures are not available or inadequate, the enforcement of laws is not going to be effective. Thus, countries in this situation would score 0. Countries also score 0 if they have the enforcement mechanisms but are not enforcing the laws (as a policy choice or because certain other policy choices make enforcement ineffective (e.g. weak fines or sentences)). However, if countries are deemed to be enforcing the laws, but not effectively because of barriers to enforcement (e.g. resource constraints) or because of delays in the implementation of policy (that is, an intellectual property Act goes into effect six months or a year later), they would score $\frac{1}{2}$. Essentially countries should score a half point if they are trying to enforce the laws (but are less successful because their capacity to enforce needs to be strengthened). For countries without enforcement problems, a score of 1 is given. Note that complaints about the lack of laws (other than enforcement provisions) are not counted in this index since the previous indexes (e.g. PAT, COPY, TMARK, and so forth) have already incorporated information about the absence of laws.

D. Sample Statistics

Table 1 presents the indexes by country, averaged over the period 1987-1995. The mean patent rights score is 2.78 (with a coefficient of variation of 0.23). The mean level for copyrights and trademarks is 3.27 and 3.26 respectively (with coefficients of variation of 0.10 and 0.17 respectively). The country with the strongest measured patent rights is the U.S. (with a level of 3.57) and the weakest is Mexico (with a score of 1.30). For copyrights, the country with the strongest measured rights is France (score of 4.0) and the weakest is again Mexico. For trademarks, the countries with strongest measured regime are France, Switzerland, and the U.K. (with scores of 4) and the weakest is Mexico, followed by Korea. For parallel import protection, 9 out of 21 countries provide it, 2 do not (Mexico and New Zealand), and the rest provide uncertain or partial protection. As for software protection, 7 countries provide protection for software (through trade secrecy, patents, and copyright): Canada, Germany, Japan, Korea, Sweden, UK, and USA. Belgium and Portugal provide the weakest software protection. Most are providing partial protection.

On measures of statutory enforcement (ENF), 7 countries provide full enforcement mechanisms, 9 near full (0.67 points), 5 near incomplete (0.33 points), and none provide incomplete mechanisms. The USTR index indicates that the vast majority of countries (i.e. 14 of them) score high on enforcement effectiveness, 6 score medium, and only 1 scores low (namely Mexico). The simple correlation between ENF and USTR is 0.24.

The rest of Table 1 shows some simple correlations among the IPR indexes. Patent rights are highly correlated with USTR; that is, countries with strong patent systems tend also to be countries that enforce laws effectively. Patent rights are positively correlated with copyrights, trademark rights,

and software rights. Software rights have almost no correlation with copyrights, though this variable was constructed using elements of copyright laws. It may be that countries that provide copyright protection for software are (in many cases) going against the grain in their national policy towards copyrights in general. Note that trademark and copyrights are highly correlated; this might have something to do with the similarity in material covered (namely, creative works, logos, business names, and so forth). Parallel import rights are more highly correlated with copyrights and trademark rights, than with patent rights and software rights. What copyrights, trademark rights, and parallel importation rights have in common is trade, sale, and marketing of goods, whereas patent rights refer to inventive activity. The USTR index of enforcement effectiveness is positively correlated with all of the other indexes, especially software and patent rights.

5. Data

A. *Manufacturing Sample*

This paper combines two OECD data sets: its *Basic Science and Technology Indicators* (1997 edition) and its manufacturing sector database (STAN). The industry by industry R&D are referred to as the BERD (i.e. Business Enterprise Research and Development). An advantage of the BERD data is that it provides separate figures for privately funded and publicly funded industrial R&D.

This sample consists of 21 countries and 18 manufacturing industries. Appendix 2 provides a list of these industries and countries in the sample. It also explains how the two data sets concord with each other (i.e. how the BERD sectoral codes match up to the sectoral codes used in the *STAN*).

From STAN, data on output (Y), labor (L), and investment in physical capital were obtained. From the science and technology database, data were obtained on privately financed business R&D (referred to here as private R&D investment, I_p) and on government-financed business R&D (referred to here as public R&D investment, I_g). Figures on output, physical capital investment, and R&D investments (public and private), were all converted to real 1990 U.S. dollars using the IMF's *Financial Statistics* data on GDP deflators and exchange rates.

Since BERD data are only available from 1987-95, we developed the data set around this time period. Moreover, there are a number of data missing in BERD. Where possible, we interpolated (where data were missing between observations) and extrapolated data (where data were missing at the end of the period). Essentially, a linear trend was fitted to the observations available.¹⁸ By doing this, we implicitly removed the influences of any business cycles.

Next, from the investment (flow) data, stocks were derived; that is, the stocks of physical capital (K), private R&D capital (R_p), and public R&D capital (R_g). The perpetual inventory method was used; for instance, let R denote the stock and I the flow. Each period, the stock is as follows:

¹⁸ In the case where only one observation was provided throughout the whole period, the same value was given for the other years.

$$(15a) \quad R(t) = I(t) + (1 - \delta) R(t-1)$$

$$(15b) \quad R(0) = \left(\frac{1 + \gamma}{\gamma + \delta} \right) I(0)$$

The initial stock $R(0)$ was obtained by backward recursive substitution of equation (15a), where γ is an historical average of the growth rate of investment (i.e. $1 + \gamma(t) = I(t)/I(t-1)$).¹⁹ A ten percent geometric depreciation rate δ was assumed.

From these data, three time-series observations were selected for each manufacturing sector (in each country): namely the averages of 1987-89, 1990-92, and 1993-1995. The IPR data corresponding to this are 1985, 1990, and 1995, since the IPR data are only available for those years. (Note that the IPR variables vary by country, not by sector.) IPR levels change very slowly so that the levels could reflect conditions for more than one year.

Next, one of the tests in this paper is to determine if IPRs interact adversely with or impede R&D spillovers (our proxy for knowledge diffusion). Thus, a measure of R&D spillovers is needed. Spillover stocks are derived, for each sector (in each country per time period), by summing up the R&D stocks of other sectors in that country (not in other countries) during that time period. Here, international R&D spillovers are not considered.²⁰ That is, for each country, the i th sector's stock of R&D spillovers at time t is:

$$(16) \quad R^*_{it} = \sum_{j \neq i}^N R_{jt}$$

The above formula applies for both public and private R&D spillover stocks.

B. Sample Statistics

Table 2 provides some summary statistics (and correlations) for the R&D-related data. Part A focuses on the sample as a whole, part B summarizes data by sector, and part C by country.

The mean rate of R&D investment (as a percentage of output) is 1.29%. The mean rate of public

¹⁹ Because the sample period is relatively short, $\gamma = 0$ is assumed if this method produces negative values for γ .

²⁰ International spillovers were considered in preliminary work, but the results are not conclusive. A problem arises because U.S. firms or sectors have the lowest (or receive the lowest) international R&D spillovers, since the bulk of world R&D is conducted by the U.S. Another issue concerning international spillovers is defining the appropriate metric of technological similarity.

R&D investment is 0.29%. The respective coefficients of variation are 1.82 and 5.10. Thus public R&D investment rates vary much more widely. There is a much larger stock of private R&D capital than of public R&D capital (about five times as much on average), and also a much larger stock of private R&D spillovers than of public R&D spillovers (about six times as much on average). The stocks per worker are all in real 1990 U.S. dollars.

There is a positive correlation (of 0.27) between public and private R&D investment rates, measured as a percentage of output, and a positive correlation (of 0.21) between public and private R&D capital per worker. The correlations are much higher between the private R&D investment rate and private R&D stock per worker (of 0.79) and between the public R&D investment rate and public R&D stock per worker (of 0.91). The private R&D investment rate and private R&D stock per worker are also each positively correlated with the stock of private R&D spillovers per worker and the stock of public R&D spillovers per worker. The correlation between patent rights and these R&D variables is positive, but the correlation between copyright and trademark rights with private R&D capital per worker is overall negative. The correlation between the IPR variables and public R&D investment rates and stocks per worker are not all highly positive nor highly negative. One issue worth investigating is whether public R&D policies and intellectual property rights policies are “substitutes” for one another. Both are alternative means of dealing with the appropriability problem with knowledge-based investments. If they are substitutes, IPR should be high where public R&D contributions are low, and vice versa. On the other hand, if they were strong complements, there would be a much higher positive correlation. Of course, these correlations are not controlling for any third factors (for example, implementation and administration costs).

Part B of Table 2 presents the R&D investment rates and stocks per worker by manufacturing sector. The heaviest concentration of public R&D is in the Aircraft sector. Its mean rate of public R&D investment per (manufacturing sector) output is 5.64%. The sector that has the highest mean rate of private R&D investment is the pharmaceutical sector (or Chemicals: Drugs), followed by the Office & Computing sector. The lowest rate of private R&D investment is in Fabricated Metals, and the lowest rate of public R&D investment is in Food and Beverages. The highest overall stock of private R&D capital per worker is in the Office & Computing industry, followed by the Chemicals: Drug sector. The sector with the smallest private R&D stock per worker is Fabricated Metals. The stock of public R&D capital per worker is highest overall in the Aircraft sector, followed by the Office & Computing industry. The stock of private R&D spillovers per worker is highest in the Ship Building industry, followed by the Petroleum industry. These sectors’ spillover stocks are high largely because, relative to their R&D, the other sectors do much more R&D (e.g. Office and Computing, Aircraft, Chemicals: Drugs, Communications, and so forth). The highest stock of public R&D spillovers is in Petroleum. The stock of private R&D spillovers and the stock of public R&D spillovers are both lowest in the Fabricated Metals industry.

Part C of Table 2 summarizes R&D statistics by country. The highest rates of private and public R&D investment are in the US (with rates of 2.77% and 1.24% respectively). The second highest rate of private R&D investment is in Sweden and second highest rate of public R&D investment is in Germany. The lowest rate of private R&D investment is in Portugal and lowest rate of public

R&D investment is in Mexico. The largest stocks of private and public R&D per worker are also in the U.S. The lowest stock of private R&D per worker is in Portugal and the lowest stock of public R&D capital per worker is in Mexico. The largest stock of private R&D spillovers per worker is in Japan and the lowest in Mexico. The largest stock of public R&D spillovers per worker is in the U.S. and the lowest in Portugal.

A comparison with Part B of this table would show that overall there is more variability in R&D investment rates and capital stock when the data are presented by country (rather than by industry). In part B, within each sector, the data vary by country; whereas in part C, for each country, the data vary by sector. Hence what drives most of the variability is the intersectoral differences (from Food and Beverages to Aircraft), which also suggests that there is a greater deal of heterogeneity across sectors rather than across countries. The empirical section which follows will fit the model (equation (14)) to the sample as a whole, as well as by sector.

6. Empirical Results

The basic results of estimating (14), where patent rights (PAT) is the measure of IPR, are presented in Table 3. The first column shows the OLS results. Public R&D capital per worker has a positive effect on private R&D capital per worker. A 1% increase in public R&D capital per worker stimulates a 0.544% increase in private R&D capital per worker. Physical capital is also seen to augment the stock of private R&D capital. A higher stock of tangible capital per worker (K/L) works, in our framework, to increase the marginal productivity of private R&D capital and thereby stimulates the accumulation of private R&D, leading to a higher stock of private R&D capital in the long run. Patent rights also positively explain the (equilibrium) stock of private R&D capital. The elasticity of private R&D capital per worker with respect to patent rights is 1.787. The model explains about 59% of the data.

In columns (2) and (3), we examine some reverse regressions, where private R&D capital might be a determinant of public R&D capital and patent rights, respectively. First, in column (2), indeed private R&D capital per worker can potentially be an important determinant of public R&D. Either the public sector is an endogenous, optimizing agent seeking to maximize social welfare (in which case it would seek to equate the discounted marginal products of public and private R&D capital, and thereby account for the positive relationship between public and private R&D) or some third factor (such as technology shocks) affect the productivity of R&D, public and private alike. Public R&D is also explained by the stock of public R&D spillovers. This could be the result of public sector research being stimulated by the public R&D conducted in other sectors or it may be the result also of some third factors; perhaps the fact that governments have national R&D objectives, more than sector-specific ones, so that a rise in public R&D for one sector is actually a rise in public R&D for all sectors as the government tries to achieve some national purpose. This latter possibility is consistent with the finding that public R&D is positively explained by government spending in general (see the GOVT variable). Public R&D is actually a small percentage of national government spending. However, an overall expansion in public spending may include some growth in public R&D spending. Note that the coefficient of 0.293 does not indicate how public R&D spending

grows in proportion to government spending in general, since the dependent variable is public R&D stock per person. The preferred interpretation, though, is that the GOVT (per worker) variable is a measure of government size, and that a larger government size is associated here with a larger long run stock of public R&D capital (because of greater public investments). But note importantly that patent rights are not a statistically significant determinant of public R&D capital per worker.

In column (3), we also see that patent rights are not determined by public R&D capital per worker. There does not seem to be a relationship between the accumulation of public R&D and the strength of patent system, as might be thought if national governments choose their mix of public R&D and levels of patent rights to achieve an overall objective with regards to how much private knowledge should be proprietary. For instance, it might have been thought that governments could either provide weak patent protection and heavy public R&D programs in their place or strong patent protection and limited public R&D involvement (thus allowing private R&D markets to take their course). On the other hand, private R&D capital per worker does significantly explain the level of patent rights, but the magnitude of the effect is small. A reason that patent rights may depend on the level of private R&D is that countries that are more intensive in research activities have more to gain from protecting research outputs. Countries where private R&D activity is relatively low may have a larger share of imitators, so that there is pressure by locals to keep patent rights weak to enable them to better imitate foreign technologies, to use proprietary knowledge, or to acquire spillover benefits. Another important determinant of patent rights is the economic freedom index (ECON).²¹ One interpretation is that patent rights measure specific (technology-related) property rights, whereas the economic freedom index measures property rights in general. Countries that protect property rights in general are more likely to protect IPRs in particular, than vice versa. Indeed where property rights are in general weak, policy authorities are more apt to be hesitant about the idea of creating intellectual property protections.

But as column (4) shows, the Hausman tests indicate that we cannot reject the null hypothesis that patent rights are exogenous (vis-a-vis private R&D capital). It might be reasonable to assume that the levels of patent rights are determined endogenously by policy authorities (in response to the nature of domestic R&D activity), it is also the case that patent reforms have taken place where domestic R&D activity is not high. One reason is that the reforms were in response to trade pressures (or incentives). Another is that countries may *expect* greater R&D to result in the future. That is, countries may have low levels of domestic R&D, but seek to raise that with higher levels of patent protection. Thus, current levels of domestic R&D activity may be an uncertain determinant of patent protection levels. The Hausman test, however, indicates a rejection of the null that public R&D is exogenous. Overall, after taking the endogeneity of public capital into account, the instrumental variables (IV) estimates indicate that (compared to the OLS findings) patent rights have a similar effect on private R&D capital, but the effect of public R&D capital on private R&D capital is higher, and the effect of physical capital per worker lower. The model explains 48% of the data.

Note that, given the coefficient estimate of PAT, the implied elasticity of the appropriability function

²¹ The data are from Gwartney and Lawson (2001).

(with respect to patent rights) must exceed one (see (13c) and expression for β_3 in (14)). This implies that the appropriability function is convex. The ability to capture revenue increases with each ‘unit’ increase in patent rights at an increasing rate. This implies that halfway measures are not very effective instruments for appropriation; starting from no protection to some halfway point of protection does not raise appropriability as much as going from that halfway point of protection to full protection of patent rights. Halfway measures leave much room still for imitation and infringement.

In Table 4, we examine the role of other (non-patent) kinds of IPRs: copyrights, trademark rights, software rights, parallel importation rights, and enforcement effectiveness. In columns (1)-(5), each of the other IPR measures is examined individually with the rest of the variables in the model. In column (6), all of the IPR variables are examined together along with the rest of the variables. The estimates are obtained by the method of IV to take into account the endogeneity of public R&D.

Copyrights and trademark rights each exert a negative influence on private R&D capital per worker. This might suggest that as instrument of intellectual property rights, they on balance restrict knowledge diffusion or the use of valuable creations, and therefore make it more difficult to build on previous research. If it were merely the case that they are not appropriate instruments for stimulating R&D (since they protect creative expressions and symbols, not inventions), the coefficient estimates (or measured effects) should have been statistically insignificant.

Software rights have a positive influence on the stock of private R&D capital. This might indicate that protection of this relatively narrow (and recent) sphere of technological inventions spurs R&D across sectors (even if they are not specifically computer-oriented) because software programs (and programming) may be important inputs of R&D production in general. Or, it might be that this variable is picking up the effects of the omitted variable, PAT (or patent rights in general).

Column (4) shows that parallel importation rights have a negative effect on R&D. It may be that parallel import protection limits the diffusion of knowledge or spillovers since it limits the diffusion of new goods to intellectual property owners or their assigned distributors. The market power effect of parallel importation rights should also be important here.

In column (5), the enforcement effectiveness variable has a predicted positive influence on R&D. Clearly, enforcement of rights and laws is an important part of having intellectual property rights.

Once patent rights are controlled for, as in column (6), the copyrights, software rights, and parallel importation rights diminish in (statistical) significance. Patent rights (PAT) and enforcement (USTR) remain significant explanatory factors. This seems intuitive - namely, that for R&D, the legal variable that matters most is patent rights and their effective enforcement. Trademark rights still exhibit a negative influence on R&D. Unlike other forms of IPRs, such as patent rights and copyrights, no “work” of knowledge per se is disclosed to the public in exchange for the trademark right. Trademarks give exclusive use of some name, symbol, or other. In other words, it may function as a type of entry barrier or market restriction or impediment without offering something

else of benefit in exchange, such as access to some know-how.

As for why software rights were also not significant once patent rights and enforcement were controlled for, it may be that adding software inventions to the list of patentable matter does not add further variability in patent rights across countries. Those that protect software are likely to protect the traditional fields (including a field that was once considered controversial, namely pharmaceuticals). Moreover, countries that do not protect invention matter like food, chemicals, etc. are not likely to consider something recent like software. (One might make a conjecture that protection for business method patents and genetic innovations will also not add much variability to the level of patent rights that the variable PAT already captures - that is, in terms revealing the strength of patent protection across countries).

It is interesting to note that *statutory* patent provisions were found to stimulate R&D, since a common criticism of the statutory patent protection variable is that it does not measure *actual* practice. Here, the evidence seems to suggest that the laws on the books can stimulate research and development. The idea may be that laws act as a *signal* - that they reveal something about the attitudes of public authorities towards the protection of intellectual property and affect confidence levels among agents to invest, say, in risky ventures like R&D.

We next pursue the question of whether IPRs can inhibit spillovers. Earlier we argued that stronger IPRs have two opposing effects: the *appropriability effect* of stronger IPRs versus any *spillover reduction effect* of tighter IPRs. However, from the coefficient of IPR, namely β_3 , we cannot separately identify the parameters of the appropriability function and spillover reduction function - namely the α and β parameters. Indeed the coefficient is a reduced form expression; it is derived to reflect the net effect, and we have found the net effect to be positive in the case of patent rights and enforcement (and negative in the case of trademark rights). Because of this inability to identify, we may not know if there exists a spillover reduction effect at all.

Thus, an alternative strategy here is to work with the interindustry spillover variables, and determine if IPRs in any way reduce the transmission of those spillovers. Consider, first, column (1) of Table 5. The stock of private R&D spillovers per worker is added to the model (equation (14)). The coefficient estimate is statistically significant at better than conventional levels, and shows the elasticity estimate to be 0.708. The stock of private R&D spillovers appears to help augment the marginal productivity of own-sector private R&D capital. But note that the patent rights variable no longer explains private R&D. In fact, an issue is whether patent protection adversely affects interindustry spillovers. Thus following interaction variable is added to the model:

$$\text{INTER1} = \ln(\text{PAT}) \times \ln(R_p^*/L),$$

where R_p^* is the stock of private R&D spillovers. This allows us to examine how the influence of private R&D spillovers on R&D -- i.e. the slope -- varies, if at all, with the level of patent rights. As shown in column (2), the coefficient estimate of this interaction term is negative indicating that patent rights reduce the (positive) impact that private R&D spillovers have on R&D capital, but the

estimate is not statistically significant at conventional levels.

The column (3) results indicate whether patent rights do the same for public R&D. That is, whether patent rights influence the relationship between public R&D and private R&D. Thus a different interaction term is added to the model:

$$\text{INTER2} = \ln(\text{PAT}) \times \ln(R_g/L),$$

where R_g is the stock of public R&D capital. The coefficient estimate of this variable is positive and statistically significant, indicating that higher levels of both public R&D and patent rights stimulate private R&D. This view suggests that the two are complementary - that they enhance the effects of the other.

In column (4), the patent rights variable is dropped since it is found not to affect private R&D, once private R&D spillovers are controlled for. In its place, the stock of public R&D spillovers is added. The finding is that this variable exerts a negative influence on private R&D (but its coefficient estimate is not statistically significant). As public R&D is already controlled for, the result suggests that private R&D is stimulated by public R&D invested in the same sector, not in other sectors. The negative (albeit weak) effect of other sectors' public R&D might be due to resource crowding out effects or resource diversion effects.

The finding that patent rights do not affect private R&D once private R&D spillovers are controlled for does not preclude the possibility that patent rights could *indirectly* affect private R&D by affecting private R&D spillovers, or by affecting private R&D activity in general (in a wide sense). This would be consistent with our earlier findings, where patent rights stimulate private R&D. Thus an auxiliary equation is estimated, one in which private R&D spillovers is a function of the stock of capital (in other sectors), the stock of public R&D in other sectors, and patent rights.

$$(14)' \quad \ln \left(\frac{R^*}{L} \right) = f \left(\ln \left(\frac{R^*}{L} \right), \ln \left(\frac{K^*}{L} \right), \ln (IPR) \right)$$

This equation is the mirror-image of (14), where the stock variables are the *sum* of other sectors' values. For example, $K_i^* = \sum_{j \neq i} K_j$. The LHS of (14)' is the *i*th sector's R&D spillovers; the RHS variables are all the sectors' values other than sector *i*'s. As for the patent rights variable, it is of course not sector-specific but a nationwide measure (and hence applies to all sectors).

Both equation (14) and (14)' are estimated jointly as a system of equations (by the method of seemingly unrelated regressions (SUR)).²² Equation (14) captures the "direct" effect of patent rights on R&D, while equation (14)' captures the indirect effect on R&D via the effect of patent rights on R&D spillovers (or on total R&D activity as a whole). Columns (5) and (6) report the results of (14)

²² The public R&D variable is again instrumented.

and (14) respectively. Again, as when equation (14) was estimated as a single equation, the patent rights variable is not an important explanatory variable when the private R&D spillover variable is included. Patent protection, however, is an important determinant of private R&D activity in general, and thus helps generate private R&D spillovers. The results in column (6) of Table 5 are the counterpart of results in column (1) of Table 5 and columns (1) and (4) of Table 3, except that the regression is run on *other sectors'* data. The auxiliary regression captures much more of the variations in the data (about 90%). This is likely because a lot of the idiosyncratic, sector-specific factors and noises are minimized, if not removed. The auxiliary regression is run over near total (or averaged) sectoral data.

In Table 6, equations (14) and (14) are estimated by sector. In addition, all of the IPR measures are included jointly. This allows us to determine which sector's R&D capital or R&D spillover capital is influenced most by which type of IPR. Part A of Table 6 reports on the direct effects of IPR (estimates of equation (14)) and Part B the indirect effects (estimates of equation (14)). It should be cautioned that at the sectoral level, there are a number of degrees of freedom lost due to a number of missing observations. At the sectoral level, it should also be noted that the variation in the data is across time and across countries.

The first rows of Parts A and B refer to all sectors pooled. The difference between these results and those in columns (5) and (6) of Table 5 is that all the IPR variables are entered together. We first discuss the results in Part A (the "direct" effects on R&D). For the sectors as a whole, public R&D capital, private R&D spillovers, and physical capital all augment the marginal productivity of private R&D capital. Of the IPR measures, only the USTR variable -- that is, the enforcement effectiveness measure -- has a direct positive and significant effect on R&D. Trademark rights have a negative (almost significant at conventional levels) effect.

Turning to the sectors, public R&D has strong positive marginal contributions to the private R&D of the Petroleum, Motor Vehicles, and Aircraft sectors. It is interesting to note that the Aircraft industry is where public R&D is most heavily involved. Public R&D has significantly negative effects on private R&D in the Wood, Chemicals (Non-Drugs), Non-Ferrous Metals, Office & Computing Equipment, Ship, and Electricals. This may signify some crowding out effects or free riding (whereby private researchers free ride on public research efforts). Physical capital has significantly positive effects on private R&D in the Wood, Drugs, Iron, Non-Ferrous, Fabricated Metals, Electricals, Motor Vehicles, and Aircraft sectors. A significantly negative effect is detected in Petroleum and Office & Computing. In Petroleum, it may be that a higher stock of tangible capital displaces R&D (intangible) capital. In the Office & Computing sector, an explanation might be that the R&D of this sector is not capital-intensive, but skilled or human capital-intensive. An expansion in capital may therefore crowd out some R&D investments.

Private R&D spillovers, in a majority of sectors, exert a significantly positive influence on private R&D. It is not important in Drugs and Electricals. One explanation might be that the R&D in these sectors is sector-specific, so that outside R&D may be less appropriate. In terms of the model in section 3, may be near zero. The negative effect of spillovers detected in the Aircraft and

Petroleum sectors may signify some free-riding effects (on the research of other sectors).

Turning to the impacts of IPRs, there appear to be a few significantly direct effects at the sectoral level. Patent rights have significantly direct positive effects on private R&D in the Wood, Chemicals (Drugs and Non-Drugs), Office & Computing, Electricals, and Shipbuilding sectors. This would indicate that the increased appropriability effects of patent rights dominate any spillover reduction effects of patent protection. On the other hand, there are significantly negative effects detected in Printing, Petroleum, indicating that the spillover reduction effects dominate on net. Copyrights have a directly significant positive effect in Chemicals (Non Drugs) and in the Aircraft sector, but have strongly negative effects in Electricals, Office & Computing, and Non-Ferrous Metals. Trademarks have a positive effect on R&D only in Electricals; this may reflect the tremendous importance of trademarks in electronic products. Otherwise, strong negative effects of trademark rights are detected in sectors such as Chemicals (Non Drugs), Non Metals, and Aircraft.

Software rights have a significantly positive effect on R&D in Chemicals (Drugs and Non-Drugs), Electricals, Aircraft, Wood Products, and the Food sector. Software may be important in the latter because some of its research relate to agricultural and biotechnological work where, as in the Chemicals sectors, software may be an important input into research and production. Software rights negatively affect private R&D in Textiles, Office & Computing. This suggests perhaps that an ‘open source’ structure may be more conducive in these sectors.

Parallel importation rights exert a positive influence in the Wood, Chemicals (Non-Drugs), Office & Computing, and Aircraft sectors. A negative influence is found in Food, Printing & Publishing, and Petroleum. The measure of enforcement effective (USTR) is a positive influence in Textiles, Wood, Chemicals (Drugs and Non-Drugs), Petroleum, and Non-Metals, and a negative influence only in Office & Computing. Some imitation (or imitative activity) may have direct positive influences on research in the latter industry.²³

It is interesting to note that software rights, patent rights, and effective enforcement were important explanatory factors for Chemicals (Drugs) research, since this sector (as we found) enjoys relatively little or no spillovers from other sectors. Thus to the extent these kinds of IPRs are important to this sector, the firms in this sector must depend on them directly, since there are no indirect channels of influence here for IPRs.

Next, we focus on the results in Part B, Table 6 (the “indirect” effects on R&D). For all sectors as a whole, the stock of private R&D spillovers is a positive function of other sectors’ capital (per own sector’s worker) and of the stock of public R&D spillovers. Among the IPR measures, patent rights, software rights, and enforcement effectiveness all contribute strongly and positively to facilitate private R&D spillovers. Parallel import rights and copyrights have a strong negative influence on

²³ In some instances, this is an argument that tighter IPRs can reduce R&D by preventing learning by imitation. While very plausible, a perhaps bigger reason for imitating in the first place is to avoid doing the R&D - to produce copies without incurring the upfront research costs.

private R&D spillovers, suggesting that they impede research knowledge spillovers.

Turning to the individual sectors, the stock of other sectors' physical capital has, in the majority of cases, a positive influence on the other sectors' private R&D capital (which constitute the stock of private R&D spillovers for the own-sector). Note that the interpretation of this variable has to be taken with care (since it is the sum of the other sector's physical capital stock); thus, this variable is significant in the Office & Computing and Petroleum sectors, whereas it was not in Part A. This is because in Part B, for the Office & Computing sector, for example, the variable (K^*/L) includes all stocks of physical capital other than that for the Office & Computing sector. The same goes for the Petroleum sector. The stock of public R&D spillovers has also in the majority of cases a strong positive effect on the stock of private R&D spillovers. This is not surprising since public R&D, for sectors as a whole, tend to have a positive effect on private R&D. (The exception is in Motor Vehicles and Petroleum, where own public R&D did have a strong positive effect on own private R&D. This might suggest that public R&D has sector-specific characteristics -- or has to be tailored -- in these sectors.)

As for the IPR variables, patent rights are an important explanatory variable except in Drugs (since there is no spillover role for patents, as this sector is weakly dependent on outside research) and in Office & Computing and Shipping, where patent rights already have a direct impact. Patent rights are also not important in Aircraft (which might be due to the large role that public R&D plays, obviating the need for strong patent incentives).

Copyrights have strong negative influences on private R&D spillovers in Textiles, Chemicals (Non Drugs), Non Metals, Iron, Fabricated Metals, and Electricals. A positive influence is found only in the Aircraft sector. (Perhaps a case study analysis could better shed light on why.) Trademarks have negative effects in the Wood, Printing, and Shipping sectors; positive effects in Petroleum and Motor Vehicles; and negligible effects in the remaining sectors. Software rights have strong negative effects on private research spillovers in Printing and Publishing (which may depend on wide dissemination to encourage R&D). Otherwise, software rights have a significantly positive effect on private R&D spillovers in several sectors: Chemical (Non-Drugs), Petroleum, Rubber, Non-Ferrous Metals, Electrical, Motor Vehicle, and Aircraft. Parallel importation rights, interestingly, have no positive effect on knowledge spillovers in any sector, but strongly negative effects in many sectors (Food, Textiles, Chemicals (Drugs and Non-Drugs), Petroleum, Rubber, Non Metals, Iron, Non Ferrous Metals, Fabricated Metals, Shipping, and Motor Vehicles). This may have to do with the very nature of parallel import rights, in limiting "distribution" to authorized agents. Finally, enforcement effectiveness is generally a positive determinant of private R&D spillovers. Again, the effect is weaker in the Chemicals sector, particularly Drugs, since R&D tends to be sector-specific and does not depend much on spillovers from other sectors. There is also a weak effect for the Office & Computing sector, but the enforcement effectiveness variable had already a direct effect. Overall, there is no significant negative effect of enforcement effectiveness on spillovers.

In sum, at the sectoral level there are both direct and indirect effects at work, though the results generally mimic the results for the sectors as a whole - that IPRs (particularly patent rights and

enforcement) matter most in stimulating R&D activity in general, thereby making intersectoral spillovers possible. But controlling for those spillovers, there is usually no additional effect of IPRs on private R&D.

7. Conclusion

As mentioned earlier, public support for R&D and intellectual property protection are both solutions to the appropriability problem in R&D knowledge accumulation (though not the only solutions). In the absence of public R&D or intellectual property rights, underinvestment (or no investment) in R&D is argued to result. Thus, public R&D and IPRs should have positive influences on R&D behavior (at least raising R&D levels above what they would be without public intervention or proprietary rights). Thus far, there has been considerable empirical work on the relationship between public and private R&D, but the empirical work on IPRs and R&D lags far behind. This paper has sought to help close that gap. The paper largely finds that both mechanisms (public R&D and IPRs, particularly patent rights and enforcement effectiveness) stimulate private R&D. Moreover, though it was not directly tested, the results do not hint at any “substitutability” between public R&D support and IPR levels (that is, where public R&D capital is high, IPR levels are weak, and vice versa). The evidence examined here favours some complementarity: that public R&D capital and patent protection each enhances the effect that the other has on private R&D capital.

A summary of the main results are as follows: first, as just mentioned, public R&D stimulates private R&D (particularly if both refer to the same sector; that is, spillovers from public R&D in other sectors matter relatively less). But the paper finds endogeneity between public and private R&D that must be taken into account. Patent protection also stimulates R&D (at an increasing rate, indicating that the underlying “appropriability function” is convex). Patent rights are found not to be determined by (nor to determine) public R&D capital, and are weakly determined by private R&D capital. Thus patent rights are on the whole treated as exogenous (to private R&D).

As for the other IPRs, copyrights and trademark rights appear to restrict R&D spillovers. Specialized forms of protection like software rights have a small value-added contribution to stimulating private R&D (when patent rights are controlled for). Parallel import rights overall have a negative effect on private R&D. The effectiveness of enforcement also matters to R&D, but even when enforcement effectiveness is controlled for, patent *statutes* (“laws on the books”) remain a significant explanatory factor for private R&D.

In terms of intersectoral spillovers, there is no evidence that patent rights impede them. Indeed they help stimulate the stock of private R&D spillovers. In turn, these spillovers are an important driving force in private R&D activities. But interestingly, once R&D spillovers are controlled for, patent rights are no longer an important determinant of private R&D. However, patent rights indirectly stimulate private R&D by positively stimulating private R&D spillovers, which in turn directly impact on private R&D.

Across individual sectors, there are of course some differences in results, but overall the general

findings or conclusions obtain at the sectoral level. The behavior of a few sectors might deviate from the overall (pooled) sample, but they are in the minority. For example, in the Chemicals (Drugs) sector, R&D spillovers are not an important determinant of private R&D. In the Chemicals (Drugs and Non-Drugs) and Office & Computing sectors, patent protection can directly affect private R&D, even though spillovers are controlled for. In Electricals, trademark rights have a strong, direct positive effect on private R&D, and in Office & Computing, software rights have a strong, direct negative effect on private R&D. These are a sample of some exceptions to the general results.

In closing, there are a couple of ways to extend this study: 1) The first is to examine copyright and trademark-related *output*. The results show that R&D is patent-sensitive, rather than copyright, trademark, or other IP-sensitive. This does not imply that copyrights, trademarks, or other non-patent IP instruments, do not matter to *other* valuable economic activity. Thus future work could explore other types of economic activity (in other words, other possible LHS variables, other than R&D) that might be more specifically a function of copyrights or trademark rights; for example, promotion, marketing, and advertising investments, education, community development, and cultural activities. 2) Another useful extension is to construct measures of *effective* intellectual property by industry. In principle, the IP laws vary by country but not by industry within a country, except in the case of coverage (for example, a country may not provide protection for biotechnological innovations, pharmaceuticals, or software). By excluding certain patentable subject matter, the laws do provide a tacit amount of preferential protection across sectors. But in theory, the IPR laws are national in scope. However, in practice, there are important inter-industry differences in the level of IPRs that firms can enjoy, and they are measurable. This matters because different kinds of inventions may require different levels of protection for the inventors to recoup their R&D costs. For some inventions, existing rights may be too weak (say chemical inventions) and for others too strong (say business methods). Another reason that firms in different industries may enjoy different *effective* levels of IP protection is that the process of procuring rights and enforcing them may differ. For example, due to differences in the complexity of technologies, firms in certain sectors may take longer to obtain a patent. The search and examination process may be more involved so that patent pendency is longer. It may also be more difficult to detect and prove infringement, and therefore enforce property rights, in certain kinds of technological fields. The ability to procure and enforce IP rights may also be function of the degree of competition in the sector.

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Appendix 1. Intellectual Property Rights - Summary of Criteria and Measurement

This appendix reviews the key criteria under each type of IPR index, and the method for scoring the strength of protection. It is likely that no one index captures the overall nature of IP protection in a region; but together the various indexes should provide a general picture.

I. PAT

This is the index of patent rights in Ginarte and Park (1997) without the enforcement category. The original index has five categories (including enforcement). The remaining four categories are:

(1) Membership in International Treaties	<u>Signatory</u>	<u>Not Signatory</u>
-- Paris Convention and Revisions	1/3	0
-- Patent Cooperation Treaty	1/3	0
-- Protection of New Varieties (UPOV)	1/3	0
(2) Coverage	<u>Available</u>	<u>Not Available</u>
-- Patentability of pharmaceuticals	1/7	0
-- Patentability of chemicals	1/7	0
-- Patentability of food	1/7	0
-- Patentability of plant and animal varieties	1/7	0
-- Patentability of surgical products	1/7	0
-- Patentability of microorganisms	1/7	0
-- Patentability of utility models	1/7	0
(3) Restrictions on Patent Rights	<u>Does Not Exist</u>	<u>Exists</u>
-- "Working" Requirements	1/3	0
-- Compulsory Licensing	1/3	0
-- Revocation of Patents	1/3	0
(4) Duration of Protection	<u>Full</u>	<u>Partial</u>
	1	$0 < f < 1$

-- where f equals the duration of protection as a *fraction* of the full (potential) duration.

FULL duration is either 20 years from the date of application or 17 years from the date of grant (for grant-based patent systems) .

II. COPY

This is an index of copyrights, based on statutory provisions:

(1) Membership in International Treaties	<u>Signatory</u>	<u>Not Signatory</u>
-- Berne Convention	1/4	0
-- Rome Treaty	1/4	0
-- Universal Copyright Convention	1/4	0
-- Phonogram Convention	1/4	0
(2) Coverage	<u>Available</u>	<u>Not Available</u>
-- Literary, Dramatic, Artistic, ...	1/5	0
-- Performance Rights	1/5	0
-- Sound Recording	1/5	0
-- Cinema	1/5	0
-- Broadcasting	1/5	0
(3) Restrictions on Copyrights	<u>Does Not Exist</u>	<u>Exists</u>
-- Limit Re-sale ("Droit-de-Suite")	1/4	0
-- Extended (Collective) Licensing Schemes	1/4	0
-- Compulsory Licensing: Government Use	1/4	0
Private Use	1/4	0
(4) Duration of Protection	<u>Full</u>	<u>Partial</u>
-- Literary, Dramatic, Artistic, ...	1/4	$0 < f < 1/4$
-- Performance	1/4	$0 < f < 1/4$
-- Sound Recording	1/4	$0 < f < 1/4$
-- Cinema	1/4	$0 < f < 1/4$

-- where f equals the duration of protection as a *fraction* of the international standard of 50 years, times 1/4. The duration of protection varies by kind of work covered. Each kind has equal weight in the overall duration score. If the duration of a work exceeds the 50 year norm, a maximum score of 1/4 is assigned.

III. TMARK

This is an index of trademark rights, based on statutory provisions:

(1) Membership in International Treaties	<u>Signatory</u>	<u>Not Signatory</u>
-- Madrid Treaty	1/3	0
-- Nice Treaty	1/3	0
-- Paris Convention	1/3	0
(2) Coverage	<u>Available</u>	<u>Not Available</u>
-- Service Marks	1/3	0
-- Certification Marks	1/3	0
-- Collective Marks	1/3	0
(3) Restrictions on Trademark Rights	<u>Does Not Exist</u>	<u>Exists</u>
-- Renewal Proof of Use	1/4	0
-- "Linking" Requirements	1/4	0
-- Restricted Licensing	1/4	0
-- Lack of Protection for Well-known Marks due to Non-Use	1/4	0
(4) Duration of Protection	<u>Full</u>	<u>Partial</u>
	1	$0 < f < 1$

-- where f equals the duration of protection as a *fraction* of the FULL duration (of ten years, the international norm).

IV. SOFT

This index measures the intellectual property protection for software. The patent and copyright indexes above do not explicitly include software in their coverage category. Hence a special index can be created for this particular type of innovation or creative expression (and/or as an extension to the indexes above).

Software rights can be protected by three sources: (a) trade secrecy protection; (b) patent; and (c) copyrights. Thus

$$\text{Software Index} = \text{Trade Secret } (x_1) + \text{Patent } (x_2) + \text{Copyright } (x_3)$$

where $x_1 = 1$ if trade secrecy protection exists (0 otherwise)

$x_2 = 1$ if patent protection exists for software, $\frac{1}{2}$ if partial provision exists, and 0 otherwise

$x_3 = 1$ if copyrights exist for software, $\frac{1}{2}$ if partial provision exists, and 0 otherwise

The information on software patent and copyright provisions is contained in *Computer Law* (Matthew Bender), Chapter 3B. Information on trade secret protection is contained in Hemnes et. al. (1992).

V. PARA

This refers to 'Parallel Import' protection for intellectual properties (e.g. books, computer programs, phonograms, videos, and so forth), as rated by the International Intellectual Property Alliance (IIPA).

The index = 1 if YES, country provides parallel import protection
 = ½ if PROBABLY YES
 = 0 if NO

VI. Enf-GP

This is the enforcement category of the Ginarte-Park (1997) index, as detached from the aggregate index. Since the same enforcement features are available for patent rights enforcement as well as for other types of intellectual property rights enforcement, it is useful to look at this category as a separate index. This index can represent the statutory provisions for enforcing IPRs.

Enforcement	<u>Available</u>	<u>Not Available</u>
-- Preliminary Injunctions	1/3	0
-- Contributory Infringement	1/3	0
-- Burden-of-Proof Reversal	1/3	0

VII. Enf-USTR

This index is a qualitative measure of the effectiveness of IPR enforcement in practice. It is based on reports filed with the U.S. Trade Representative (USTR) which document the experience of IP enforcement in countries outside the U.S.

The reports describe complaints, if any, about enforcement procedures and/or about the failure of the proper authorities to carry out the laws on the books. The failure to enforce may be due to some inability on the part of the authorities to carry out those laws or due to a conscious policy choice. Absence of substantive laws (other than enforcement provisions) are already incorporated in the previous indexes, and thus complaints about the lack of substantive laws are not incorporated here. Thus, the index is given by:

Enf-USTR = 0 if enforcement measures are not available or inadequate (e.g. weak deterrents);
 ½ if enforcement measures are available but not effectively carried out (due to lag in policy implementation or resource barriers);
 1 otherwise.

Appendix 2. Manufacturing Industry Sample

(A) List of Countries in the Sample:

Australia	Korea
Austria	Mexico
Belgium	Netherlands
Canada	New Zealand
Denmark	Norway
Finland	Portugal
France	Spain
Germany	Sweden
Greece	United Kingdom
Italy	United States
Japan	

(B) List of Industries in the Sample (for each country):

<u>Sector:</u>	<i>Line Number from:</i>	
	<u>STAN</u> <u>Database</u>	<u>DSTI</u> <u>Database</u>
1. Food, Beverages, Tobacco	3100	04
2. Textiles, Apparel, Leather	3200	07
3. Wood Products & Furniture	3300	12 & 40
4. Printing & Paper Products	3400	13 & 14
5. Chemicals (Non-Drugs)	3512x	18
6. Chemicals (Drugs)	3522	19
7. Petroleum	3534A	16
8. Rubber & Plastics	3556A	20
9. Non-metallic Mineral Products	3600	21
10. Metals: Iron & Steel	3710	23
11. Metals: Non-Ferrous Metals	3720	24
12. Fabricated Metal Products	3800	25
13. Office & Computing Equipment	3825	28
14. Radio, TV, & Communication Equipment	3832	32
15. Electrical Apparatus (excl. Communication Equip.)	383X	29
16. Shipbuilding	3841	36
17. Motor Vehicles	3843	34
18. Aircraft	3845	37

Notes:

STAN is the OECD Industrial Activity Database

DSTI is the OECD Science & Technology Indicators Database.

Table 1. Intellectual Property Rights - Descriptive Statistics

	PAT	COPY	TMARK	SOFT	PARA	USTR	ENF
Austral	2.99	2.87	3.08	2.5	0.5	1	0.33
Austria	3.24	3.33	3.42	2	0.5	1	1
Belgium	2.90	3.50	3.67	1.5	0.5	0.5	1
Canada	2.09	2.96	3.08	3	1	1	0.67
Denmark	3.23	3.67	3.67	2.5	0.5	1	0.67
Finland	2.28	3.28	3.67	2.5	0.5	1	0.67
France	2.90	4.00	4.00	2.5	1	1	1
Germany	3.04	3.54	3.17	3	1	1	0.67
Greece	1.65	3.10	2.83	2	1	0.5	0.67
Italy	3.05	3.42	3.42	2.5	1	0.5	1
Japan	3.27	3.10	2.50	3	0.5	1	0.67
Korea	2.94	2.80	2.25	3	0.5	0.5	1
Mexico	1.30	2.66	2.00	2	0	0	0.33
Netherl.	3.52	3.10	3.67	2.5	0.5	1	0.67
New Zeal.	2.32	2.82	2.45	2.5	0	1	1
Norway	2.62	3.00	3.42	2.5	1	1	0.67
Portugal	1.65	3.50	3.67	1.5	0.5	0.5	0.33
Spain	3.29	3.65	3.75	2.5	1	0.5	0.33
Sweden	3.23	3.67	3.67	3	0.5	1	0.67
U.K.	3.24	3.47	4.00	3	1	1	0.33
U.S.A.	3.57	3.35	3.17	3	1	1	1
Mean	2.78	3.27	3.26	2.50	0.67	0.81	0.70
Std Dev	0.63	0.34	0.56	0.46	0.32	0.29	0.25
Min	1.30	2.66	2.00	1.5	0	0	0.33
Max	3.57	4	4	3	1	1	1

Sample Correlations (1987-95)

	PAT	COPY	TMARK	SOFT	PARA	USTR	ENF
PAT	1						
COPY	0.36	1					
TMARK	0.41	0.81	1				
SOFT	0.44	-0.03	-0.11	1			
PARA	0.26	0.45	0.47	0.32	1		
USTR	0.56	0.22	0.38	0.54	0.21	1	
ENF	0.35	0.11	-0.05	0.16	0.07	0.24	1

Definitions:

PAT refers to patent rights, COPY to copyrights, TMARK to trademark rights, SOFT to software intellectual property rights, PARA to parallel import protection, USTR to a qualitative index of effective enforcement (based on reports to the USTR), and ENF to statutory enforcement rights.

Table 2: R&D Data -- Descriptive Statistics*(A) Total Sample (21 countries, 18 manufacturing sectors):*

Variables:	Mean	Std. Dev	Min	Max
Private R&D Investment to Output (%) ----- Ip/Y	1.29	2.35	0.0002	17.21
Public R&D Investment to Output (%) ----- Ig/Y	0.29	1.48	0.0001	21.05
Private R&D Stock to Labor ----- RDp/L	21393	55386	1.255	988194
Public R&D Stock to Labor----- RDg/L	4680	24509	0	270833
Private Spillover R&D Stock to Labor----- RDp*/L	459411	1015952	454	16500000
Public Spillover R&D Stock to Labor----- RDg*/L	69400	178045	0	1496495
Physical Capital Stock to Labor----- K/L	93005	168423	1415	2714487

	Correlations between:						
	Ip/Y	Ig/Y	RDp/L	RDg/L	RDp*/L	RDg*/L	K/L
Ip/Y	1						
Ig/Y	0.27	1					
RDp/L	0.79	0.17	1				
RDg/L	0.25	0.91	0.21	1			
RDp*/L	0.14	0.07	0.31	0.08	1		
RDg*/L	0.25	0.00	0.44	0.01	0.51	1	
K/L	-0.03	-0.04	0.13	-0.02	0.16	0.09	1
PAT	0.16	0.08	0.22	0.12	0.22	0.31	0.01
COPY	0.03	0.06	-0.01	0.07	0.03	0.17	-0.08
TMARK	-0.03	-0.01	-0.08	-0.01	-0.09	0.03	-0.05
SOFT	0.19	0.11	0.22	0.12	0.12	0.21	0.01
PARA	0.12	0.13	0.14	0.13	-0.01	0.24	0.06
USTR	0.16	0.03	0.18	0.05	0.16	0.13	0.07
ENF	0.05	0.05	0.05	0.06	0.15	0.24	0.07

Notes:

Private R&D refers to privately-financed Business Enterprise R&D (in real 1990 US\$)

Public R&D refers to government-financed Business Enterprise R&D (in real 1990 US\$)

Y denotes output, K capital (both in real 1990 US\$), and L labor

n/a indicates data not available

For other variable definitions, see Notes to Table 1

Table 2 ...cont'd/

(B) By Manufacturing Sector:

	<i>Mean/Std. Dev (in italics)</i>						
	Ip/Y	Ig/Y	RDp/L	RDg/L	RDp*/L	RDg*/L	K/L
Food, Beverages	0.18	0.005	2846	85	70708	15009	64333
	<i>0.14</i>	<i>0.004</i>	<i>2231</i>	<i>77</i>	<i>62884</i>	<i>31055</i>	<i>20656</i>
Textiles, Apparel	0.17	0.011	1162	82	108478	11590	30521
	<i>0.13</i>	<i>0.016</i>	<i>1103</i>	<i>123</i>	<i>134498</i>	<i>15418</i>	<i>13478</i>
Wood Products	0.096	0.015	777	131	144133	14202	35964
	<i>0.069</i>	<i>0.024</i>	<i>708</i>	<i>196</i>	<i>116487</i>	<i>20067</i>	<i>19335</i>
Printing and Paper	0.16	0.007	1997	78	94733	10092	85449
	<i>0.15</i>	<i>0.009</i>	<i>2353</i>	<i>132</i>	<i>77883</i>	<i>13448</i>	<i>39536</i>
Chemicals (Non-drug	1.64	0.05	35683	1064	192875	43986	119040
	<i>1.17</i>	<i>0.04</i>	<i>28638</i>	<i>1071</i>	<i>135568</i>	<i>72100</i>	<i>70008</i>
Chemical (Drug)	7.13	0.17	80687	1498	673215	155795	76494
	<i>4.72</i>	<i>0.21</i>	<i>62429</i>	<i>1413</i>	<i>479801</i>	<i>272491</i>	<i>55101</i>
Petroleum	0.35	0.02	37206	2604	1968170	308704	489207
	<i>0.39</i>	<i>0.022</i>	<i>41682</i>	<i>3504</i>	<i>1570815</i>	<i>460669</i>	<i>519524</i>
Rubber & Plastics	0.61	0.024	5933	175	198466	21695	52299
	<i>0.49</i>	<i>0.035</i>	<i>5257</i>	<i>235</i>	<i>156376</i>	<i>21460</i>	<i>25105</i>
Non-metallic Minerals	0.51	0.022	5752	197	220967	45668	76604
	<i>0.43</i>	<i>0.025</i>	<i>5329</i>	<i>169</i>	<i>198863</i>	<i>95517</i>	<i>37118</i>
Metals: Iron & Steel	0.44	0.023	8335	392	255678	61111	136478
	<i>0.34</i>	<i>0.022</i>	<i>8154</i>	<i>402</i>	<i>220699</i>	<i>125172</i>	<i>147128</i>
Metals: Non-Ferrous	0.43	0.048	10123	598	617230	131795	117517
	<i>0.37</i>	<i>0.104</i>	<i>9775</i>	<i>302</i>	<i>479530</i>	<i>215987</i>	<i>100186</i>
Fabricated Metals	0.078	0.009	689	87	20369	3996	49651
	<i>0.087</i>	<i>0.009</i>	<i>481</i>	<i>86</i>	<i>13680</i>	<i>6791</i>	<i>20545</i>
Office & Computing	4.18	0.43	134618	19449	1042862	202520	68078
	<i>2.84</i>	<i>0.83</i>	<i>188983</i>	<i>53807</i>	<i>879484</i>	<i>210217</i>	<i>45106</i>
Communications	3.72	0.67	39061	9121	201517	57906	51290
	<i>2.29</i>	<i>0.87</i>	<i>25183</i>	<i>12086</i>	<i>112295</i>	<i>82331</i>	<i>26078</i>
Electricals	1.53	0.19	16065	1563	175263	43678	46873
	<i>0.93</i>	<i>0.38</i>	<i>10528</i>	<i>2063</i>	<i>150302</i>	<i>84678</i>	<i>28442</i>
Shipbuilding	0.81	0.34	10519	3806	2283049	264474	29359

	0.86	0.58	13366	7527	3281185	390921	23541
Motor Vehicles	1.85	0.12	24741	1682	171152	16895	71274
	1.43	0.25	23545	3620	170863	13299	40759
Aircraft	3.29	5.64	34865	79512	1168711	50098	65606
	2.42	5.43	26871	81156	1653132	33654	35552

Table 2 ...cont'd/

(C) *By Country:*

Mean/Std. Dev (in italics)

	Ip/Y	Ig/Y	RDp/L	RDg/L	RDp*/L	RDg*/L	K/L
Australia	0.94	0.023	6304	289	149668	2290	53017
	<i>1.25</i>	<i>0.046</i>	<i>8351</i>	<i>557</i>	<i>186783</i>	<i>2321</i>	<i>53328</i>
Austria	1.29	0.051	17709	595	823359	28911	79502
	<i>1.87</i>	<i>0.059</i>	<i>21158</i>	<i>683</i>	<i>2508193</i>	<i>91179</i>	<i>49093</i>
Belgium	1.06	0.042	6637	202	426250	9833	79049
	<i>1.75</i>	<i>0.091</i>	<i>5676</i>	<i>198</i>	<i>614812</i>	<i>9126</i>	<i>38748</i>
Canada	1.15	0.21	17317	2551	245508	37002	98805
	<i>1.69</i>	<i>0.71</i>	<i>20984</i>	<i>7463</i>	<i>230688</i>	<i>36725</i>	<i>107133</i>
Denmark	1.51	0.024	11493	233	107546	2847	66912
	<i>2.45</i>	<i>0.032</i>	<i>20682</i>	<i>394</i>	<i>130397</i>	<i>3642</i>	<i>52852</i>
Finland	0.47	0.025	8262	219	111771	3228	110116
	<i>0.35</i>	<i>0.037</i>	<i>7168</i>	<i>287</i>	<i>125554</i>	<i>4038</i>	<i>83603</i>
France	1.42	0.38	24490	7490	657414	186723	131280
	<i>1.91</i>	<i>1.37</i>	<i>30116</i>	<i>22606</i>	<i>1096639</i>	<i>303791</i>	<i>220172</i>
Germany	1.71	0.94	26122	11184	889782	79864	70283
	<i>2.64</i>	<i>3.69</i>	<i>36547</i>	<i>41258</i>	<i>1234906</i>	<i>113192</i>	<i>52013</i>
Greece	0.18	0.03	44834	13957	217153	27749	89813
	<i>0.21</i>	<i>0.04</i>	<i>188455</i>	<i>55443</i>	<i>783595</i>	<i>90829</i>	<i>146448</i>
Italy	1.69	0.55	13602	5001	238006	71981	83114
	<i>2.99</i>	<i>1.56</i>	<i>23522</i>	<i>15438</i>	<i>279956</i>	<i>87118</i>	<i>50459</i>
Japan	1.53	0.12	31414	1919	1004122	20638	126203
	<i>1.83</i>	<i>0.45</i>	<i>37995</i>	<i>5817</i>	<i>1546383</i>	<i>32828</i>	<i>131582</i>
Korea	n/a	n/a	n/a	n/a	n/a	n/a	84099
	n/a	n/a	n/a	n/a	n/a	n/a	<i>99040</i>
Mexico	0.15	0.0003	476	1.1	8812		
	<i>0.29</i>	<i>0.0002</i>	<i>688</i>	<i>0.104</i>	<i>9966</i>	n/a	n/a
Netherlands	n/a	n/a	n/a	n/a	n/a	n/a	112454
	n/a	n/a	n/a	n/a	n/a	n/a	<i>126795</i>
New Zealand	n/a	n/a	n/a	n/a	n/a	n/a	117307
	n/a	n/a	n/a	n/a	n/a	n/a	<i>268571</i>
Norway	1.28	0.072	26297	1211	371512	18639	213611

	2.18	0.14	53693	2630	384658	20262	505107
Portugal	0.06	0.04	366	66	6466	324	35609
	0.07	0.14	586	229	9442	473	21458
Spain	1.22	0.32	14603	3143	221489	28237	43620
	2.41	1.41	38061	9899	325713	44549	45412
Sweden	2.22	0.19	37039	2032	828796	53355	103374
	4.19	0.71	66209	5192	1097639	60331	94152
U.K.	1.59	0.39	20482	4465	573448	66901	54444
	3.17	1.03	37588	11102	910958	63244	52613
U.S.A.	2.77	1.24	55799	23203	765914	491357	60948
	3.37	3.38	61054	66098	667373	420644	42335

Table 3. R&D and Patents -- Basic Results

	<i>Dependent Variable:</i>			
	----- OLS -----			-- IV --
	(1)	(2)	(3)	(4)
	RD _p /L	RD _g /L	PAT	RD _p /L
Constant	-3.527 (0.822)	-3.271 (0.508)	0.569 (0.061)	-3.304 (0.941)
RD _g /L	0.544 (0.025)		0.0005 (0.0036)	0.762 (0.045)
K/L	0.524 (0.066)			0.446 (0.082)
PAT	1.787 (0.298)	-0.480 (0.358)		1.752 (0.299)
RD _p /L		0.654 (0.043)	0.026 (0.005)	
RD _g */L		0.142 (0.046)		
GOVT		0.293 (0.074)		
ECON			0.297 (0.028)	
Adj. R ²	0.59	0.55	0.30	0.48
No. of Observ.	640	661	664	640
² (4) ^I [p-value]				1.28 [0.87]
² (4) ^{II} [p-value]				89.6 [0.00]

Notes: RD_p, RD_g denote the stocks of *private* and *public* R&D capital (in real 1990 U.S. dollars) respectively. L denotes labor, K capital, PAT patent rights index, and IV instrumental variables estimation. All variables are in natural logarithms. Heteroskedastic-consistent standard errors are in parentheses. ²(4)^I is the Hausman test-statistic (with four degrees of freedom) for testing the null hypothesis of the exogeneity of PAT and ²(4)^{II} the Hausman test-statistic for testing the null hypothesis of the exogeneity of RD_g/L. The instrument for PAT is the index of economic freedom (ECON), and the instruments for RD_g/L are the stock of public R&D spillovers (from other sectors), i.e. RD_g*/L, and national government spending per worker (GOVT).

Table 4. R&D and other Intellectual Property Rights

<i>Dependent Variable: RD_p/L</i>						
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	0.091 (1.169)	1.102 (1.057)	-2.610 (0.937)	-1.636 (0.855)	-0.375 (0.861)	0.686 (1.314)
RD _g /L	0.825 (0.039)	0.829 (0.038)	0.761 (0.044)	0.898 (0.044)	0.769 (0.043)	0.804 (0.050)
K/L	0.456 (0.084)	0.432 (0.081)	0.482 (0.083)	0.439 (0.081)	0.404 (0.082)	0.337 (0.078)
PAT						1.179 (0.332)
COPY	-1.208 (0.589)					0.544 (0.884)
TMARK		-1.827 (0.465)				-1.997 (0.638)
SOFT			1.258 (0.312)			-0.338 (0.435)
PARA				-0.847 (0.179)		-0.388 (0.206)
USTR					1.458 (0.192)	1.210 (0.289)
Adj. R ²	0.46	0.47	0.47	0.47	0.50	0.53
No. of Obs.	640	640	640	640	640	640

Notes: Estimation is by IV (instrumental variables), where the instruments for RD_g/L include the stock of public R&D spillovers (RD_g^{*}/L) and national government spending per worker (GOVT). All variables are in natural logarithms. See previous tables for variable definitions. Heteroskedastic-consistent standard errors are in parentheses.

Table 5. R&D Spillovers and Patent Rights

	<i>Dependent Variable:</i>					
	(1) RD _p /L	(2) RD _p /L	(3) RD _p /L	(4) RD _p /L	(5) RD _p /L	(6) RD _p [*] /L
Constant	-5.328 (0.942)	-6.376 (1.581)	-1.786 (0.863)	-5.214 (0.939)	-7.397 (0.989)	-12.76 (0.389)
RD _g /L	0.115 (0.086)	0.125 (0.089)	0.409 (0.047)	0.227 (0.096)	0.096 (0.089)	
K/L	0.405 (0.080)	0.410 (0.079)	0.412 (0.071)	0.394 (0.080)	0.403 (0.068)	
K [*] /L						0.645 (0.018)
PAT	-0.082 (0.391)	0.808 (1.306)	0.771 (0.301)		-0.135 (0.373)	2.404 (0.118)
RD _p [*] /L	0.708 (0.091)	0.822 (0.174)		0.757 (0.075)	0.729 (0.088)	
RD _g [*] /L				-0.127 (0.068)		0.321 (0.014)
INTER1		-0.102 (0.143)				
INTER2			0.301 (0.021)			
² (1) [p-value]						0.076 [0.783]
Adj. R ²	0.53	0.53	0.62	0.53	0.53	0.91
No. of Obs.	640	640	640	640	640	640
Estimation Method:	IV	IV	IV	IV	SUR	SUR

Notes: Equations 1 - 4 are estimated by instrumental variables, where the instruments for RD_g/L include the stock of public R&D spillovers (RD_g^{*}/L) and national government spending per worker (GOVT). Equations 5 and 6 are estimated jointly by the method of seemingly unrelated regressions; ²(1) is the Breusch-Pagan test statistic (with one degree of freedom) for testing the null hypothesis of independent equations. INTER1 is an *interaction variable* equal to the product of the log of PAT and the log of RD_p^{*}/L, and INTER2 is the product of the log of PAT and the log of RD_g/L. All variables are in natural logarithms. See previous tables for variable definitions. Heteroskedastic-consistent standard errors are in parentheses.

Table 6: Interindustry R&D Spillovers - Direct and Indirect Effects - SUR Estimates**A. Direct Effect -- Dependent Variable (RD_p/L)**

	<u>Const.</u>	<u>RD_g/L</u>	<u>K/L</u>	<u>RD_p^*/L</u>	<u>PAT</u>	<u>COPY</u>	<u>TMARK</u>	<u>SOFT</u>	<u>PARA</u>	<u>USTR</u>	<u>R²</u>
All Sectors N=640	-3.552 (1.489)	0.416 (0.116)	0.348 (0.069)	0.406 (0.112)	0.362 (0.411)	0.268 (0.863)	-1.258 (0.656)	-0.503 (0.445)	-0.163 (0.215)	0.946 (0.269)	0.55
Food... N=51	-3.686 (3.401)	0.302 (0.170)	-0.106 (0.191)	0.294 (0.112)	0.565 (0.498)	-0.971 (0.988)	1.051 (0.722)	1.379 (0.502)	-1.478 (0.261)	0.097 (0.309)	0.82
Textiles.. N=48	5.781 (6.244)	-0.123 (0.284)	-0.406 (0.347)	1.055 (0.176)	-1.877 (0.741)	0.439 (1.548)	-1.658 (1.292)	-3.484 (0.899)	-0.522 (0.379)	1.213 (0.494)	0.81
Wood... N=39	-29.801 (6.358)	-1.526 (0.415)	0.434 (0.228)	0.463 (0.303)	2.390 (0.896)	3.541 (2.157)	-1.696 (1.920)	0.489 (0.936)	1.333 (0.559)	3.209 (0.696)	0.78
Printing... N=42	-0.458 (8.952)	0.215 (0.628)	0.302 (0.337)	0.761 (0.303)	-5.338 (1.579)	-5.114 (3.091)	4.812 (2.624)	0.919 (1.304)	-2.422 (0.959)	0.629 (0.784)	0.38
NonDrug... N=31	-12.275 (1.408)	-0.566 (0.091)	0.042 (0.089)	0.722 (0.098)	1.178 (0.368)	2.982 (0.461)	-1.401 (0.335)	1.794 (0.269)	1.022 (0.163)	0.977 (0.194)	0.95
Drug.. N=29	-17.726 (2.104)	-0.335 (0.269)	0.264 (0.104)	0.067 (0.145)	3.353 (0.781)	-0.716 (1.128)	1.559 (0.961)	4.555 (0.650)	0.421 (0.537)	1.118 (0.477)	0.89
Petro... N=26	21.416 (4.312)	2.811 (0.303)	-0.312 (0.078)	-0.609 (0.294)	-1.910 (0.648)	0.447 (1.202)	-1.275 (0.880)	0.322 (1.270)	-3.509 (0.776)	1.206 (0.310)	0.95
Rubber.. N=43	-3.342 (5.653)	-0.418 (0.272)	0.032 (0.287)	1.297 (0.233)	-0.008 (0.706)	2.437 (1.343)	-2.127 (1.271)	-3.917 (0.785)	0.467 (0.313)	0.046 (0.478)	0.73
NonMetals.. N=51	-3.785 (4.387)	0.008 (0.259)	0.063 (0.248)	0.717 (0.191)	1.246 (0.797)	2.236 (1.692)	-3.120 (1.221)	-1.609 (0.841)	-0.316 (0.449)	1.295 (0.571)	0.80
Iron ... N=43	-11.229 (2.766)	-0.403 (0.158)	0.474 (0.182)	1.231 (0.194)	-0.151 (0.663)	0.196 (1.718)	0.158 (0.910)	-1.026 (0.927)	-0.706 (0.372)	0.578 (0.469)	0.85

Table 6 cont'd ../**A. Direct Effect -- Dependent Variable (RD_p/L)**

	<u>Const.</u>	<u>RD_g/L</u>	<u>K/L</u>	<u>RD_p*/L</u>	<u>PAT</u>	<u>COPY</u>	<u>TMARK</u>	<u>SOFT</u>	<u>PARA</u>	<u>USTR</u>	<u>R²</u>
NonFerrous.. N=40	-18.374 (3.733)	-0.662 (0.209)	0.933 (0.107)	0.840 (0.179)	0.645 (0.732)	-4.475 (1.507)	2.129 (0.891)	0.545 (0.796)	-0.247 (0.371)	0.176 (0.407)	0.88
Fab Metals.. N=45	-9.965 (5.294)	0.380 (0.243)	0.845 (0.241)	0.117 (0.169)	1.341 (0.723)	-1.639 (1.552)	-1.144 (1.165)	-0.641 (0.651)	-0.338 (0.432)	-0.479 (0.428)	0.65
Office .. N=30	1.655 (1.124)	-0.518 (0.185)	-0.520 (0.102)	1.946 (0.238)	1.680 (0.374)	-4.653 (0.829)	3.112 (0.724)	-1.821 (0.719)	1.303 (0.293)	-0.558 (0.252)	0.95
Commun.. N=14	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	
Electr... N=33	-19.245 (2.623)	-0.459 (0.179)	0.488 (0.175)	0.006 (0.382)	2.689 (0.854)	-4.172 (1.085)	3.345 (0.791)	3.562 (0.803)	0.150 (0.485)	0.490 (0.487)	0.85
Ship.. N=26	-27.717 (5.423)	-1.355 (0.391)	0.194 (0.237)	1.414 (0.389)	5.970 (1.781)	1.299 (3.216)	2.483 (2.635)	-1.374 (1.131)	-0.201 (0.331)	-0.325 (0.503)	0.85
Motor Veh.. N=28	-13.987 (2.366)	1.197 (0.297)	0.252 (0.126)	-0.273 (0.221)	-0.209 (0.819)	10.595 (1.229)	-1.733 (0.955)	5.848 (1.097)	-3.753 (0.669)	-0.445 (0.402)	0.84
Aircraft.. N=24	-19.936 (4.129)	1.381 (0.309)	0.467 (0.235)	-1.655 (0.253)	-0.669 (0.411)	12.161 (1.298)	-4.637 (0.743)	12.127 (1.321)	3.258 (0.346)	-0.721 (0.297)	0.95

Table 6 cont'd ../

B. Indirect Effect -- Dependent Variable (RD_p^*/L)

	<u>Const.</u>	<u>K*/L</u>	<u>RD_g^*/L</u>	<u>PAT</u>	<u>COPY</u>	<u>TMARK</u>	<u>SOFT</u>	<u>PARA</u>	<u>USTR</u>	<u>$\chi^2(1)$</u>	<u>R²</u>
All Sectors N =640	-7.587 (0.587)	0.482 (0.019)	0.485 (0.019)	1.417 (0.118)	-1.420 (0.253)	0.249 (0.177)	0.409 (0.127)	-0.831 (0.074)	0.379 (0.073)	0.101	0.94
Food... N =51	-2.763 (2.494)	0.187 (0.136)	0.555 (0.063)	1.563 (0.407)	-1.722 (0.927)	0.106 (0.649)	0.090 (0.438)	-0.996 (0.261)	0.496 (0.275)	0.004	0.92
Textiles.. N =48	-3.397 (2.166)	0.383 (0.097)	0.658 (0.065)	1.176 (0.396)	-2.177 (0.858)	0.051 (0.591)	-0.204 (0.408)	-1.123 (0.239)	0.363 (0.296)	3.096	0.95
Wood... N=39	-1.766 (2.326)	0.051 (0.127)	0.411 (0.066)	1.383 (0.312)	1.212 (0.901)	-2.410 (0.727)	0.572 (0.386)	-0.172 (0.239)	1.265 (0.253)	0.001	0.95
Printing... N=42	6.158 (4.286)	-0.240 (0.258)	0.644 (0.079)	2.126 (0.515)	0.331 (1.583)	-3.344 (1.553)	-1.362 (0.675)	-0.418 (0.395)	0.780 (0.302)	1.343	0.90
NonDrug... N=31	-8.385 (2.054)	0.409 (0.109)	0.336 (0.066)	1.732 (0.409)	-2.139 (0.607)	0.996 (0.553)	1.076 (0.371)	-0.832 (0.275)	0.052 (0.294)	0.084	0.92
Drug.. N=29	-9.230 (4.407)	0.691 (0.188)	0.552 (0.087)	0.110 (0.502)	-1.117 (0.719)	0.401 (1.178)	-0.015 (0.692)	-1.300 (0.514)	-0.702 (0.513)	0.981	0.90
Petro... N=26	-13.141 (1.374)	0.347 (0.075)	0.098 (0.069)	1.496 (0.347)	0.774 (0.821)	1.703 (0.523)	2.991 (0.412)	-1.361 (0.259)	0.571 (0.173)	0.941	0.95
Rubber.. N=43	-4.432 (2.129)	0.249 (0.095)	0.425 (0.067)	1.347 (0.319)	-1.384 (0.730)	0.002 (0.469)	0.855 (0.389)	-0.504 (0.213)	0.842 (0.225)	0.349	0.94
NonMetals.. N=51	-4.604 (2.587)	0.372 (0.127)	0.571 (0.064)	1.187 (0.414)	-2.010 (0.908)	0.471 (0.629)	0.002 (0.431)	-1.063 (0.259)	0.434 (0.313)	0.708	0.93
Iron ... N=43	3.695 (2.499)	-0.169 (0.115)	0.418 (0.064)	1.328 (0.349)	-2.915 (0.897)	0.033 (0.515)	0.620 (0.371)	-0.620 (0.226)	0.446 (0.226)	1.510	0.91

Table 6 cont'd ../

B. Indirect Effect -- Dependent Variable (RD_p^*/L)

	<u>Const.</u>	<u>K*/L</u>	<u>RD_g^*/L</u>	<u>PAT</u>	<u>COPY</u>	<u>TMARK</u>	<u>SOFT</u>	<u>PARA</u>	<u>USTR</u>	<u>$\chi^2(1)$</u>	<u>R²</u>
NonFerrous.. N=40	-0.728 (2.202)	-0.196 (0.138)	0.487 (0.068)	1.294 (0.395)	0.423 (0.984)	0.319 (0.534)	1.659 (0.493)	-1.777 (0.305)	-0.166 (0.251)	0.135	0.93
Fab Metals.. N=45	-0.726 (3.343)	0.065 (0.182)	0.529 (0.084)	1.444 (0.463)	-2.433 (1.229)	0.354 (0.693)	-0.197 (0.499)	-0.961 (0.313)	0.564 (0.299)	1.380	0.85
Office .. N=30	-8.485 (2.624)	0.452 (0.087)	0.209 (0.090)	0.332 (0.421)	0.763 (0.844)	-0.469 (0.705)	0.903 (0.782)	0.384 (0.375)	0.278 (0.231)	0.039	0.87
Commun.. N=14	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	n/a (n/a)	
Electr.. N=33	-3.138 (1.230)	0.125 (0.058)	0.257 (0.042)	1.249 (0.245)	-1.788 (0.407)	0.221 (0.302)	0.917 (0.245)	0.081 (0.215)	0.873 (0.133)	2.143	0.95
Ship.. N=26	-1.522 (4.005)	0.420 (0.169)	0.496 (0.094)	-0.529 (0.642)	1.409 (0.901)	-3.332 (1.081)	-1.003 (0.719)	-0.669 (0.158)	0.719 (0.255)	1.801	0.95
Motor Veh.. N=28	-14.649 (3.973)	0.695 (0.124)	0.228 (0.135)	0.728 (0.471)	-0.553 (0.999)	1.732 (0.668)	1.406 (0.720)	-0.964 (0.429)	0.458 (0.245)	0.370	0.90
Aircraft.. N=24	-8.046 (3.053)	0.464 (0.086)	0.888 (0.113)	-0.945 (0.477)	1.836 (0.540)	0.599 (0.690)	1.709 (0.745)	-0.369 (0.237)	1.115 (0.188)	0.080	0.95

Notes:

-- The direct and indirect equations are estimated as a system via the method of seemingly unrelated regressions (SUR).

-- N denotes the number of observations, asterisk (*) the sum of other sectors' values, and $\chi^2(1)$ the Breusch-Pagan test statistic (with one degree of freedom) for testing the null hypothesis of independent equations. Heteroskedastic-consistent standard errors are in parentheses.

-- n/a indicates estimate not possible or reliable due to insufficient observations.

-- See previous tables for variable definitions.